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MOST IMPORTANT MCQ'S WITH ANSWER KEY AND SOLUTIONS FOR MDCAT 2021

MOTION AND FORCE (UNIT#01)

Q.1 Acceleration of 1.5 ms^{-2} expressed in km hr^{-2} is:

A) 324
B) 5400
C) 5.4
D) 19440

Q.2 When a body is accelerated:

A) Its velocity always changes
B) Its direction always changes
C) its speed always change
D) its speed may or may not change

Q.3 Two railway truck of masses m and 3m moving towards each other in opposite direction with velocities 2v and v respectively these truck collide and stuck together what is velocities if trucks after collision:

A) V/4
B) V
C) V/2
D) 5V/4

Q.4 The area between the velocity and the time axis in the velocity time graph is numerically equal to the:

A) Speed of the object
B) Momentum of the object
C) acceleration of the object
D) distance covered by the object

Q.5 An object with constant speed:

A) Is not accelerated
B) Is always accelerated
C) might be accelerated
D) also has a constant

Q.6 A train of 150 m length is going towards north direction at a speed of 10 m/s. A parrot flies at a speed of 5 m/s towards south direction parallel to the railway track. The time taken by the parrot to cross the train is equal to:

A) 12 s
B) 15 s
C) 8 s
D) 10 s

$v = 10 + 5$

$s = v\,t$ or $t = \frac{s}{v}$

$\Rightarrow \quad t = \frac{150}{15} = 10$ sec

Q.7 The distance covered by a body in time t starting from rest is:

A) $at^2 / 2$
B) $a^2t / 2$
C) vt
D) at^2

$S = v_it + \frac{1}{2}at^2$ (Here $v_i = 0$)

$S = \frac{at^2}{2}$

Q.8 The range of a projectile when launched at an angle of 15° with the horizontal is 1.5 m. Its range, when launched at 45° with the same speed is:

A) 3.0 m
B) 6.0 km
C) 1.5 m
D) 0.75 km

$R = \frac{v_i^2 \sin 2\theta}{g} \rightarrow 1.5 = \frac{v_i^2 \sin 2(15)}{9.8} \Rightarrow v_i = 5.5\ ms^{-1}$

Therefore: $R' = \frac{(5.5^2)\sin 2(45)}{9.8} = 3$ m

Q.9 A body starting from rest covers a distance of 0.45 km and acquires a velocity of 300 km/hr. Its acceleration will be:

A) 0.092 m^{-2}
B) 7.71 m^{-2}
C) 0.5 m^{-2}
D) 0.15 m^{-2}

$v_i = 0$, $v_f = 300\ kmh^{-1} = 83.3ms^{-1}$

$S = 0.45$ km $= 450$ m

Using the relation $v_f^2 - v_i^2 = 2aS \rightarrow (83.3)^2 - 0 = 2a\,(450)$

$a = 7.71\ ms^{-2}$

Q.10 If the slope of a velocity-time graph gradually decreases then the body is said to be moving with:

A) Positive acceleration
B) Uniform acceleration
C) decreasing acceleration
D) negative acceleration

Q.11 The velocity of a projectile is maximum:

A) at middle of the height
C) at one-fourth of the height

B) just before striking the ground and point of projection
D) none of these

Q.12 A train takes 1 hour to go from one station to the other. It travels at a speed of 30 km h^{-1} for the first half hour and at the speed of 40 km h^{-1} for the next half hour. The average speed of the train is:
A) 45 km h^{-1}
B) 40 km h^{-1}
C) 35 km h^{-1}
D) none of these

Q.13 A body is thrown vertically upward with initial velocity 9.8 ms^{-1}. It will reach the height:
A) 19.8 m
B) 9.8 m
C) 29.4 m
D) 4.9 m

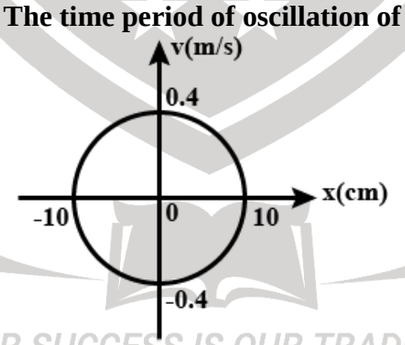
Q.14 The velocity of a particle at an instant is 10 m/s and after 5s the velocity of the particle is 20 m/s. The velocity 3s before in m/s is:
A) 8
B) 6
C) 4
D) 7

$a = \frac{v}{t}$ or $v = at$ Here, $a = \frac{20-10}{5} = 2\ ms^{-2}$
And $t = 5 - 3 = 2$ sec $\Rightarrow$ $v = (2)(2) = 4$

Q.15 Which of the following statement is true
A) Action reaction are equal in magnitude and balance each other
B) Action reaction are equal and opposite but never balance
C) Action reaction are equal and in same direction
D) None of these

Q.16 A man is at rest in the middle of a horizontal plane of perfectly smooth ice. He can move himself to the shore by making use of Newton's
A) All the laws
B) First law
C) third law
D) second law

Q.17 The plot of velocity (v) versus displacement (x) of a particle executing simple harmonic motion is shown in figure. The time period of oscillation of particle is



A) $\frac{\pi}{2}s$
B) πs
C) $2\pi s$
D) $3\pi s$

Q.18 Two bodies are thrown vertically upwards with their initial speeds in the ratio 2:3. The ratio of the maximum heights attained by them is
A) 4:9
B) $\sqrt{2}:\sqrt{3}$
C) 1:1
D) 2:3

Q.19 A body goes from A to B with a velocity of 40 kmh^{-1} and return from B to A with a velocity of 60 kmh^{-1}. What is the average velocity of the body during the whole journey?
A) 45kmh^{-1}
B) 50 kmh^{-1}
C) 55 kmh^{-1}
D) zero

Q.20 The ratio of displacement along diameter and total distance along circle:
A) $1: \pi$
B) $\pi:4$
C) $\pi:1$
D) $\pi:2$

Q.21 If a body moving in a circular path with constant speed then the
A) Velocity and acceleration are perpendicular to each other
B) Velocity and acceleration are parallel to each other
C) Velocity and acceleration are anti-parallel
D) Both have zero magnitude

Q.22 When you drop a ball it accelerates downward at 9.8 m/s^2. If you instead throw it downward, then its acceleration immediately after leaving your hand, assuming no air resistance, is:
A) 9.8 m/s^2
B) Less than 9.8 m/s^2
C) More than 9.8 m/s^2
D) Depends on throwing speed

Q.23 Three particles of masses 1 g, 2g and 3g are dropped from different heights of 1m, 2m and 3m respectively, Now,
A) Particle of mass 1 g has maximum acceleration
B) Particle of mass 2 g has maximum acceleration
C) Particle of mass 3 g has maximum acceleration
D) All the particles have equal acceleration

Q.24 A particle starts with initial velocity u and retardation a. It will reach the point of start in time:
A) $\frac{u}{a}$
B) $\frac{u}{a^2}$
C) $\frac{u^2}{a}$
D) $\frac{2u}{a}$

Q.25 Two bodies of masses m_1 and m_2 fall from heights h_1 and h_2 respectively. The ratio of their velocities, when they hit the ground, is:
A) $\frac{h_1}{h_2}$
B) $\sqrt{\frac{h_1}{h_2}}$
C) $\frac{m_1 h_1}{m_2 h_2}$
D) $\frac{{h_1}^2}{{h_2}^2}$

Q.26 Nine five-rupee coins are put on top of each other on a table. Each coin has a mass m kg. The reaction of the 5th coin (counted from bottom) on 6th coin is:
A) 0 mg
B) 2 mg
C) 4 mg
D) 5 mg

Q.27 The correct statement from the following is:
A) A body having zero velocity will not necessarily have zero acceleration
B) A body having zero velocity will necessarily have zero acceleration
C) A body having uniform speed can have only uniform acceleration.
D) A body having non-uniform velocity will have zero acceleration

Q.28 A football kicked vertically up reaches a height h and comes down to the starting point in 4 s. The value of h is:
A) 9.8 m
B) 2 x 9.8 m
C) 3 x 9.8 m
D) 4 x 9.8 m

Q.29 In a game of tug of wars, a condition of equilibrium exists. Both the teams pull the rope with a force of 10^4 N. The tension in the rope is:
A) 10^4 N
B) 10^8 N
C) 0
D) 2 x 10^4 N

Q.30 A U^{238} nucleus decays by emitting an alpha particle of speed v m/s. The recoil speed of the residual nucleus is:
A) $\frac{v}{4}$ ms^{-1}
B) - $\frac{4}{238}$ v ms^{-1}
C) - $\frac{4}{234}$ v ms^{-1}
D) $\frac{4}{238}$ v ms^{-1}

Q.31 A particle of mass "m" moving with velocity v strikes a stationary particle of mass "2m" and sticks to it. The speed of the system will be:
A) v/2
B) 2v
C) v/3
D) 3v

Q.32 A body, initially at rest, explodes into two masses M_1 and M_2 that move apart with speed v_1 and v_2 respectively. What is the ratio $\frac{v_1}{v_2}$?
A) $\frac{M_1}{M_2}$
B) $\frac{M_2}{M_1}$
C) $\left(\frac{M_1}{M_2}\right)^{\frac{1}{2}}$
D) $\left(\frac{M_2}{M_1}\right)^{\frac{1}{2}}$

Q.33 The force needed to fracture a covered skin with hair is_____ times than cranium:
A) 10 times
B) 2 times
C) 5 times
D) 50 times

Q.34 A ball sticking to the wall is a perfectly inelastic collision
A) The velocity of the ball after the collision is zero
B) Its kinetic energy is then zero.
C) All of the kinetic energy has been lost.
D) All of these

Q.35 An aero plane flying at a constant speed releases a bomb. As he bomb moves away from the aero plane, it will:

A) Always be vertically below the aero plane only if the aero plane was flying horizontally
B) Always be vertically below the aero plane only if the aero plane was flying at an angle of 45° to the horizontal
C) Always be vertically below the aero plane
D) Gradually fall behind the aero plane if the aero plane was flying horizontally

Solution:
At all times, the horizontal component of the velocity of the bomb will be the same as the horizontal component of the velocity of the aero plane. So, the horizontal displacements would remain the same at all times.

Q.36 Two bodies are projected at angles θ and (90°- θ) to the horizontal with the same speed. The ratio of their times of flight is:

A) Sin θ: 1
B) cos θ: 1
C) Sin θ: cos θ
D) cos θ: sin θ

Solution:
$T_1 = \frac{2v\sin\theta}{g}$
$T_2 = \frac{2v\sin(90^o-\theta)}{g}$ or $T_2 = \frac{2v\cos\theta}{g}$
Dividing $\frac{T_1}{T_2} = \frac{\sin\theta}{\cos\theta}$

Q.37 A projectile thrown with a velocity v and angle θ has a range R on the surface of earth. For same v and θ, its range on the surface of moon will be:

A) 36 R
B) R / 36
C) R / 6
D) 6 R

Solution:
$R \propto \frac{1}{g}$. is reduced by a factor of 6 on the moon.

Q.38 If the velocity of the projection of a projectile is trebled, then its maximum range will be:

A) Quadruple
B) Nine times
C) six times
D) eight times

Solution:
Max. Range $\propto v^2$

Q.39 If the range of a projectile is R, then its K.E is maximum after covering a distance equal to:

A) R / 4
B) R
C) R / 2
D) 3R / 4

Q.40 For which pair ranges of projectile are same

A) 50°, 45°
C) 50°, 40°
B) 80°, 20°
D) All of these

Q.41 An aeroplane flying horizontally with a speed of 360 km/h release a bomb at a height of 490 m from the ground. When will the bomb strike the ground?

A) 8 s
B) 6 s
C) 7 s
D) 10 s

Solution:
$t = \sqrt{\frac{2 \times 490}{9.8}}\ s = \sqrt{\frac{2 \times 49 \times 100}{9.8}}\ s = \sqrt{100}\ s = 10\ s$

WORK AND ENERGY (UNIT#02)

Q.1 Water from a stream is falling on the blades of a turbine at the rate of 100 kgs^{-1}. If the height of the stream is 100m, then the power delivered to the turbine is:

A) 1 kW
B) 100 W
C) 10 kW
D) 100 Kw

$\frac{m}{t} = 100$ as $P = \frac{W}{t} = \frac{mgh}{t} = \left(\frac{m}{t}\right)(gh)$

Q.1 A moving body need not have:

A) kinetic energy
B) Potential energy
C) momentum
D) velocity

Q.2 A light body A and a heavy body B have equal kinetic energies of translation. Then:

A) A has larger momentum than B
B) B has larger momentum than A
C) A and B have same momentum
D) nothing can be said about their momentum relationship unless actual masses are known

Q.3 One watt-hour equals:

A) 3.6×10^3 J
B) 3.6×10^3 cal
C) 3.6×10^6 J
D) 3.6×10^6 cal

Q.4 Energy required to accelerate a car from 10 ms^{-1} to 20 ms^{-1} compared with that to accelerate is from 0 to 10 ms^{-1} is:

A) Twice
B) three times
C) four times
D) same

$$\frac{E_2}{E_1} = \frac{\frac{1}{2} m\,[(20)^2-(10)^2]}{\frac{1}{2} m\,[(10)^2-0^2]} = 3$$

Q.5 An engine pumps up 100 kg of water through a height of 10 m in 5s. Given that efficiency of engine is 60%. If g=10ms^{-2} , the power of this engine is:

A) 3.3KW
B) 0.033KW
C) 0.20KW
D) 33KW

P_{OUT} =Power given to the water is =100×10×10/5=2000W

P_{IN} =Power of the engine= $\frac{P_{OUT}}{\eta} \times 100$ =2000×100/60=3.3 kW

Q.42 An automobile travelling with a speed of 60 km/h, can brake to stop with in a distance of 20 m. If the car is going twice as fast i.e, 120 km/h, the stopping distance will be:

A) 60 m
B) 40 m
C) 20 m
D) 80 m

Q.6 A body of mass 2kg is thrown up vertically with a kinetic energy of 490 J. If g = 9.8 ms^{-2}, the height at which the kinetic energy becomes half its original value:

A) 10 m
B) 12.5 m
C) 25 m
D) 50 m

mgh = $\frac{1}{2}$(K.E.) or h = $\frac{1}{2}$(490 J)$\left(\frac{1}{2\text{ kg x }9.8\text{ ms}^{-2}}\right)$ → h = 12.5 m

Q.7 The retarding force required to reduce velocity of a 3 kg body from 0.75 ms^{-1} to 0.25ms^{-1} in 0.02 seconds will be:

A) 100 N
B) 75 N
C) 50 N
D) 25 N

$v_f = v_i + at$ And F = ma

Q.8 A body is dropped from a height of 20 m and rebounds to a height of 10 m. The loss of energy is:

A) 10%
B) 45%
C) 50%
D) 75%

Q.9 A bullet is fired from a riffle. If the riffle recoils freely, then the kinetic energy of the rifle will be.

A) same as that of bullet
B) more than that of bullet
C) less than that of bullet
D) none of these

Q.10 Two masses of 1 g and 4 g are moving with equal kinetic energies. The ratio of the magnitude of their linear momenta is:

A) 4 : 1
B) $\sqrt{2}$: 1
C) 1 : 2
D) 1 : 16

If E = K.E. and P = momentum. Then: $E = \frac{1}{2}mv^2 = \frac{1}{2}\frac{m^2v^2}{m} = \frac{1}{2}\frac{P^2}{m}$

Therefore $E_1 = E_2$ or $\frac{P_1}{m} = \frac{P_1}{m_2}$

That is: $\frac{P_1}{P_2} = \sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$

Q.11 A child on a swing is 1 m above the ground at the lowest point and 6 m above the ground at the highest point. The horizontal speed of the child at the lowest point of the swing is approximately:

A) 8 ms^{-1}
B) 10 ms^{-1}
C) 12 ms^{-1}
D) 14 ms^{-1}

As the swing comes down, the gravitational potential energy is converted into K.E. so

$\frac{1}{2}mv^2 = mgh$

$v = \sqrt{2gh}$

$v = \sqrt{2 \times 10 \times 5}$

v = 10 ms^{-1}

Q.12 If angle between force and displacement is less than 90° then work done is
A) Positive
B) Negative
C) Zero
D) Virtual

Q.13 A bullet of mass 10 g hits a target and penetrates 2 cm into it. If the average resistance offered by the target is 100 N, then the velocity with which the bullet hits the target is:
A) 10 ms^{-1}
B) $10\sqrt{2}$ ms^{-1}
C) 20 ms^{-1}
D) $20\sqrt{2}$ ms^{-1}

$W = \frac{1}{2}mv^2$ or $Fd = \frac{1}{2}mv^2$

Q.14 A body of mass 5 kg, initially at rest, is moved by a force of 2 N on a smooth horizontal surface. The work done by the force in 10 sec is:
A) 20 J
B) 30 J
C) 40 J
D) 60 J

W = FS
To find S using the equation
$S = v_i t + \frac{1}{2}at^2$ (Here, $a = \frac{F}{m}$)

Q.15 The work done is said to be negative if:
A) $\theta = 0$
B) $\theta = \infty$
C) $\theta > 90^o$
D) $\theta < 90^o$

Q.16 The area under the force displacement curve represents:
A) Work done by the force
B) acceleration produced by the force
C) Impulse of the force
D) motion produced by the force

Q.17 Work done against the force of frictions the example of:
A) Positive work
B) negative work
C) Zero work
D) maximum work

Q.18 The force which can do no work on the body on which it acts is:
A) Frictional force
B) elastic force
C) Gravitational force
D) centripetal force

Q.19 The work done by the gravitational force is negative if the displacement of the object and the gravitational force are:
A) in same direction
B) in opposite direction
C) perpendicular
D) making an angle of 60º

Q.20 A field in which the work done in moving a body along a closed path is zero is called:
A) electromagnetic field
B) nuclear field
C) conservative field
D) resistive field

Q.21 The field in which the work done is independent of the path followed is called:
A) nuclear field
B) conservative field
C) resistive field
D) electromagnetic field

Q.22 Which of the following is not a conservative force?
A) Electric force
B) elastic spring force
C) Frictional force
D) gravitational force

Q.23 The rate at which work is done is called:
A) Energy
B) work
C) Power
D) momentum

Q.24 At which angle between force and displacement work done by force is half than its maximum
A) 30°
C) 45°
B) 60°
D) 90°

Q.25 If $\theta > 90^o$ work is said to be
A) Negative
C) positive
B) Zero
D) none of these

Q.26 A force of 10N acts on a body and body moves 10m distance perpendicular to it. Work done by the force on the body is
A) 10J
C) 100J
B) zero
D) infinite

Q.27 Which one is bigger unit of work
A) 10J
C) kilo joule
B) Mega joule
D) kilo watt hour

Q.28 The field or space around the earth in which the gravitational forces acts on a body is called
A) Gravitational field
B) Magnetic field
C) electric field
D) idea filed

Q.29 The average power and instantaneous power becomes equal if the work is done at
A) Variable rate
B) Average rate
C) uniform rate
D) high rate

Q.30 A brick of mass 2.0 kg is dropped from a rest position 5.0 m above the ground. What is its velocity at a height of 3.0 m above the ground?
A) 6.3m/s
B) 5.3m/s
C) 8.3m/s
D) 9.3m/s

Q.31 The K.E of a body depends upon
A) mass of body
B) speed of body
C) displacement of the body
D) both A and B

Q.32 Absolute P.E of a body of mass one kg over the surface of the earth is
A) $-\frac{GMm}{R}$
B) $\frac{GMm}{R}$
C) $-\frac{GM}{R}$
D) $-\frac{GRm}{M}$

Q.33 1000 N force is required to lift a hook and 10000 N force is requires to lift a load slowly. Find power required to lift hook with load with speed v=0.5m/sec
A) 5kw
B) 5.5kw
C) 1.5kw
D) 4.5kw

ROTATIONAL AND CIRCULAR MOTION (UNIT#03)

Q.1 1 degree is equal to _____ radians:
A) π/180
B) 180/π
C) 2π/180
D) 1/57.3

Q.2 When a body moves in a circle, the angle between its linear acceleration 'a' and angular acceleration 'α' is always.
A) 180°
B) 90°
C) 0°
D) 45°

Q.3 A body is rotating clockwise with decreasing angular velocity. Its angular acceleration is directed
A) Into the plane of paper
B) Out of the plane of paper
C) Along the radius
D) along the tangent to the circle

Q.4 A wheel rotates with a constant acceleration of 2 rad/s^2. If the wheel starts from rest, the number of revolution it makes in first ten seconds will be approximately
A) 8
B) 24
C) 16
D) 32

$$\theta = 0 + \frac{1}{2}(2)(10)^2$$
$$\Rightarrow \theta = 100\,\text{rad}$$
$$\Rightarrow \theta = \frac{100}{2\pi}\ \text{revolutions} \approx 15.9\ \text{revolutions}$$

Q.5 Two bodies of equal masses are moving on two circular paths of radii in ratio 1:2 with same angular velocity. The ratio of centripetal forces acting on those bodies is:
A) 1:2
B) 1:4
C) 2; 1
D) 4:1

Q.6 Two cars of masses m_1 and m_2 are moving along the circular path of radius r_1 and r_2. They take one round in the same time. The ratio of angular velocity of the two cars will be:
A) $m_1:m_2$
B) 1:1
C) $r_1:r_2$
D) $m_1:m_2$: $m_2:r_2$

Q.7 Angular speed of daily rotation of each is:
A) $\frac{\pi}{24}$ rad/h
B) $\frac{\pi}{12}$ rad/h
C) $\frac{\pi}{6}$ rad/h
D) $\frac{\pi}{2}$ rad/h

Q.8 A ball tied to a string moves in a vertical circle. Tension at the highest point will be
A) Maximum
B) Minimum
C) zero
D) positive

Q.9 The ratio of distance to the angular displacement for a body moving in circle of radius 5 m is:

A) 5 m
B) 5π m
C) $\frac{5}{\pi}$ m
D) $\frac{1}{5}$ m

Q.10 The angular acceleration of a body experiences torque, directed:

A) Towards the center of the circle
B) Away from the center of the circle
C) along the tangent
D) along the axis of rotation

Q.11 The ratio of angular velocities of the hour hand and minute hand of a watch is ————

A) 1:1
B) 720:1
C) 43200: 1
D) 1: 12

Hour hand of watch takes 12 hours and minute hand 1 hour of one complete rotation.

$\omega_1 ; \frac{2\pi}{T_1} = \frac{2\pi}{12} = \frac{\pi}{6}$

$\omega_2 ; \frac{2\pi}{T_2} = \frac{2\pi}{1} = \frac{2\pi}{1}$

$\frac{\omega_1}{\omega_2} = \frac{\pi}{6} \times \frac{1}{2\pi} = \frac{1}{12} = 1:12$

Q.12 The ratio of magnitude of angular velocity of hour- hand of a watch to that of earth's rotation about its own axis is:

A) 3:1
B) 1:3
C) 2:1
D) 1:4

Hour hand of watch takes 12 Hours and Earth 24 Hours for one complete rotation.

For which:

$\omega_1 ; \frac{2\pi}{T_1} = \frac{2\pi}{12} = \frac{\pi}{6}$

For earth:

$\omega_2 ; \frac{2\pi}{T_2} = \frac{2\pi}{24} = \frac{\pi}{12}$

$\frac{\omega_1}{\omega_2} = \frac{\pi}{6} \times \frac{12}{\pi} = 2:1$

Q.13 If a racing bike moves with a velocity of 10 ms"^{1} on a circle of radius 20 m then its angular velocity is:

A) 0.5 rad s^{-1}
B) 10 rad s^{-1}
C) 20 rad s^{-1}
D) 200 rad s^{-1}

$V = r\,\omega$

$\omega = \frac{v}{r} = \frac{10}{20} = 0.5$ rad s^{-1}

Q.14 A car is turning around a corner at 10 m/sec is travelling along an arc of circle. If value of centripetal acceleration is 10 m/sec^2, then radius of circle is:

A) 1 m
B) 5 m
C) 10 m
D) 15 m

$a_c = \frac{v^2}{r}$

$\Rightarrow r = \frac{v^2}{a_c} = \frac{100}{10} = 10$ m

Q.15 The angular speed of a particle moving along a circle is 4 rad/sec, its time period is:

A) $\frac{\pi}{2}$ sec
B) $\frac{\pi}{4}$ sec
C) $\frac{\pi}{12}$ sec
D) $\frac{3\pi}{2}$ sec

$\boldsymbol{\omega = \frac{2\pi}{T}}$

$\Rightarrow \mathbf{T = \frac{2\pi}{\omega} = \frac{2\pi}{4} = \frac{\pi}{2}s}$

Q.16 An object of mass 2 kg rotates at constant speed in a horizontal circle of radius 5 m, the time for one completer revolution is 3 sec. What is the magnitude of the resultant force acting on the object?

A) $\frac{4\pi^2}{9}$ N
B) $\frac{40\pi^2}{9}$ N
C) $\frac{100\pi^2}{9}$ N
D) $\frac{400\pi^2}{9}$ N

$F = m\,r\,\omega^2$

$\therefore \quad F = (2)\,(5)\left(\frac{2\pi}{3}\right)^2 \qquad \omega = \frac{2\pi}{T} = \frac{2\pi}{3} = \frac{40\pi^2}{9}$

Q.17 Joule-second is a unit of:
A) energy
B) momentum
C) power
D) angular momentum

Q.18 If a body moves with a constant speed in a circle:
A) no work is done on it
B) no force acts on it
C) no acceleration is produced in it
D) its velocity remains constant

Q.19 For a particle moving in uniform circular motion:
A) velocity is transverse and acceleration is radial
B) velocity is radial and acceleration is transverse
C) both velocity and acceleration are radial
D) both velocity and acceleration are transverse

Q.20 A wheel rotates about an axis passing through the center and perpendicular to the plane with slowly increasing angular speed. Thus it has:
A) radial velocity and radial acceleration
B) tangential velocity and radial acceleration
C) tangential velocity and tangential acceleration
D) tangential velocity but acceleration having both components

Q.21 A body is moving in a circle with a constant speed. It has:
A) a constant velocity
B) a constant acceleration
C) a velocity of constant magnitude and a variable acceleration
D) both velocity and acceleration of constant magnitude

Q.22 The work done by the centripetal force F when a body completes one revolution around a circle of radius R is:
A) $2\pi RF$
B) 2RF
C) RF
D) zero

Q.23 If the distance between two masses id doubled, the gravitational attraction between them:
A) Is doubled
B) becomes four times
C) Is reduced to half
D) is reduced to a quarter

Q.24 A car moving on a horizontal road may be thrown out of the road in taking a turn:
A) by the gravitational force
B) due to the lack of proper centripetal force
C) due to the rolling frictional force between the tyre and the rode
D) by the reaction of the ground

Q.25 The intensity of the earth's gravitational field is maximum at:
A) the center of the earth
B) the equator
C) the poles
D) none of these

Q.26 Which of the following statements about the motion of an earth-satellite is/are correct?
A) The satellite is always accelerating towards the earth
B) There is no force acting on the satellite
C) The acceleration and the velocity of the satellite are in the same direction
D) The satellite must fall back to the earth when its fuel is consumed

Q.27 An artificial satellite orbiting around the earth does not fall down because the attraction of the earth:
A) Vanishes at such distances
B) is balanced by the attraction of the moon
C) is balanced by the centrifugal force
D) Provides the necessary acceleration for its motion in a curved path

Q.28 A satellite is revolving around the earth in a circular orbit with a uniform speed v. If the gravitational force suddenly disappears, the satellite will:
A) Continue to move in the same orbit with speed v
B) move tangentially to the orbit with speed v
C) Move away from the earth normally to the orbit
D) Fall down on the earth

Q.29 A particle starts from rest with an acceleration 2 rads^{-2} in a circle of radius 2 m. Find its linear speed after 6 s:
A) 12 ms^{-1}
B) 24 ms^{-1}
C) 4 ms^{-1}
D) none of these

Q.30 A particle moves in a circular path of radius r. In half the period of revolution its displacement and distance covered are:

A) 2r, πr
B) r, πr
C) 2r, 2πr
D) $r\sqrt{2}$, πr

Q.31 Angular momentum of a body under a central force is:

A) Zero
B) maximum
C) Minimum
D) constant

Q.32 A car of mass 1000 kg moves on a circular road with a speed of $20ms^{-1}$. Its direction changes by 90º after traveling 628 m on the road. The centripetal force acting on the car is:

A) 500 N
B) 750 N
C) 1000 N
D) 1500 N

$\frac{2\pi r}{4} = 628$

or $r = 400$ m

Centripetal force = $\frac{mv^2}{r} = \frac{1000 \times (20)^2}{400} = 1000$ N

Q.33 If R is the radius of the earth and g is the acceleration due to gravity on the surface of the earth, the mean density of the earth is:

A) $\frac{3\pi R}{4gG}$
B) $\frac{4\pi G}{4gG}$
C) $\frac{4RG}{3\pi g}$
D) $\frac{3g}{4\pi RG}$

$\rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi R^3}$

$g = \frac{GM}{R^2} \Rightarrow \rho = \frac{3g}{4\pi RG}$

$M = \frac{gR^2}{G}$

Q.34 If 'g' is the acceleration due to gravity at the surface of the earth , then the energy required to launch a satellite of mass m from the surface of the earth into a circular orbit at an altitude of 2R, R being the radius of the earth, is:

A) $\frac{mgR}{6}$
B) $\frac{mgR}{3}$
C) $\frac{2mgR}{3}$
D) $\frac{5mgR}{6}$

$E_i = -\frac{GMM}{R}$, $E_f = -\frac{GMM}{2(3R)}$

Energy required = E_f □ E_i

$= \frac{GMM}{R} - \frac{GMM}{6R}$

$= \frac{5}{6}$ mgR

Q.35 A point on the rim of wheel moves 0.2 m when the wheel turns through an angle of 0.25 rad. The radius of wheel is:

A) 0.65 m
B) 0.125 m
C) 0.8 m
D) 0.34 m

$S = r\theta$

$r = \frac{S}{\theta}$

Q.36 The angular speed of the minute hand of the clock in degrees per second is:

A) 1.0
B) 0.1
C) 0.001
D) none of these

Q.37 The angular speed for daily rotation of earth in rad s^{-1} is:

A) 2π
B) π
C) 7.3×10^2 $rads^{-1}$
D) 7.3×10^{-5} $rads^{-1}$

Q.38 A man mass 100 kg is standing in an elevator. The net force man when elevator is going up with acceleration of 4 ms^{-2} would be:

A) 908 N
B) 490 N
C) 100 N
D) 1380 N

F = T - W

ma = T - mg

T = m (a + g)

T = 100(4 + 9.8)
T = 1380 N

Q.39 What will be the duration of the day and night (in hour) if the diameter of the earth is suddenly reduced to half its original value, the mass remanding constant:

A) 12 B) 6
C) 3 D) 2

$I_1\omega_1 = I_2\omega_2$

or $I_1\frac{2\pi}{T_1} = I_2\frac{2\pi}{T_2}$ ……..(i)

Also $I_2 = M\left(\frac{D}{2}\right)^2 = \frac{I_1}{4}$

And $T_1 = 24$ h

So we can write eq. (i) as:

Q.40 The rate of change of angular momentum of a body is:

A) the applied force B) the momentum of inertia
C) the applied torque D) impulsive force

WAVES (UNIT#04)

Q.1 Progressive waves of frequency 300 Hz are superimposed to produce a system of stationary waves in which adjacent crests are 1.5 m apart. What is the speed of the progressive waves?

A) 100 ms^{-1} C) 450 ms^{-1}
B) 200 ms^{-1} D) 900 ms^{-1}

Q.2 The spectrum of a star's light is measured and the wavelength of one of the lines as the sodium's line is found to be 589nm. The same line has the wavelength of 497nm when observed in the laboratory. This means the star is:

A) Moving away from the earth B) moving towards the earth
C) stationary D) revolving around the planet

Q.3 Mathematical formula of maximum velocity (v_o) for a body executing simple harmonic motion is:

A) $v_o = \omega x_o$ C) $v_o = v\sqrt{1-\frac{x^2}{x_o^2}}$
B) $v_o = \frac{1}{\omega}\sqrt{x_o^2 - x^2}$ D) $v_o = \omega\sqrt{x_o^2 - x^2}$

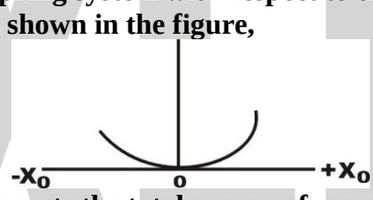
Q.4 What should be the ratio of kinetic energy to the total energy for simple harmonic oscillator

A) $\left(1-\frac{x^2}{x_o^2}\right)$ C) $(x_o^2 - x^2)$
B) 1 D) $1/2x^2$

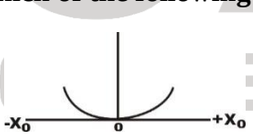
Q.5 What should be the length fo simple pendulum whose period is 6.28 second at place where g = 10ms^{-2}

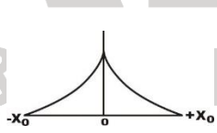
A) 9.8 m C) 6.28 m
B) 10.8 m D) 10 m

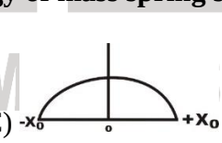
Q.6 Potential energy of a mass spring system with respect to displacement during simple harmonic motions (SHM) is shown in the figure,

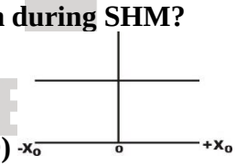


Which of the following represents the total energy of mass spring system during SHM?

A) 

B) 

C) 

D) 

Q.7 One degree Celsius ———— in temperature produces approximately 0.61 m/sec ———— the speed of sound.

A) fall, decreased C) fall, increase
B) Rise, decrease D) Rise, increase

Q.8 An observer move with velocity 'v_o' toward a stationary source, then the number of waves received in one second is

A) $f' = f\left(\frac{v}{v+v_o}\right)$
B) $f' = f\left(\frac{v}{v-v_o}\right)$
C) $f' = f\left(\frac{v+v_o}{v}\right)$
D) $f' = f\left(\frac{v-v_o}{v}\right)$

Q.9 The acceleration of simple pendulum while it moves towards the extreme position is:

A) Towards mean position
B) Towards center of earth
C) Along the direction of motion
D) randomly changes with vibration

Q.10 The relation for restoring force for simple pendulum is:

A) – mgcos θ
B) + mgcos θ
C) + mgcos θ
D) – mgsin θ

Q.11 The ratio of the kinetic energy to the potential energy at $x = \frac{x_0}{2}$:

A) 1:1
B) 3:1
C) 2:1
D) 1:3

Q.12 Potential energy of a simple harmonic oscillator is maximum at:

A) Mean position
B) Extreme position
C) In between mean and extreme position
D) All of these

Q.13 The time taken by a simple harmonic oscillator to move from mean to half of extreme is:

A) $\frac{T}{4}$
B) $\frac{T}{6}$
C) $\frac{T}{8}$
D) $\frac{T}{12}$

Q.14 A body executing simple harmonic motion has a maximum acceleration equal to 24 m/s^2 and maximum velocity equal to 16 m/s. The amplitude of the simple harmonic motion is:

A) 32/3 m
B) 1024/9m
C) 3/32 m
D) 64/9 m

Q.15 The phase difference between the acceleration and velocity of a particle executing SHM is

A) 90° B) 180° C) 360° D) zero

Q.16 The graph between velocity and acceleration of a body performing S.H.M is

A) A straight-line
B) Ellipse
C) circle
D) parabola

Q.17 The graph between potential energy and amplitude in case of SHM is a

A) Parabola
B) Sine curve
C) Straight line
D) Circle

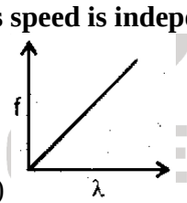
Q.18 A body of mass M is suspended from a spring of force constant K and mass m. The time period of vertical oscillation is:

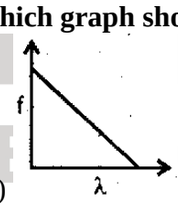
A) $2\pi\sqrt{\frac{M}{K}}$
B) $2\pi\sqrt{\frac{m}{K}}$
C) $2\pi\sqrt{\frac{M+m}{K}}$
D) $2\pi\sqrt{\frac{M+m/3}{K}}$

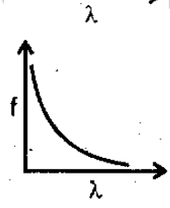
Q.19 The displacement of a particle in simple harmonic motion in one time period is:

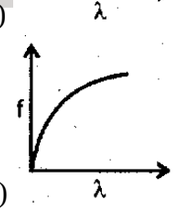
A) A
B) 2 A
C) 4 A
D) zero

Q.20 A sound wave of frequency f and wavelength X travels through air. It may be assumed that its speed is independent of the frequency. Which graph shows the variation of f with λ?

A) 

C) 

B) 

D) 

Q.21 The speed of a wave in a medium is 760 ms^{-1}. If 3600 waves pass through a point in that medium in 2 minutes, its wavelength is:
A) 13.8 m
B) 41.5 m
C) 25.3 m
D) 57.2 m

Q.22 The speed of sound in air would become double than its speed at 10 ℃ at a temperature of:
A) 313 ℃
B) 586 ℃
C) 859 ℃
D) 899 ℃

Q.23 If the speed of sound in air at a given pressure is V, then by increasing the pressure to double, the new speed becomes;
A) 0.5 V
B) V
C) 2 V
D) 4 V

Q.24 The velocity of sound at 0 ℃ is 332 ms^{-1}, the velocity of sound at 10 ℃ will be:
A) 337.1 ms^{-1}
B) 338.1 ms^{-1}
C) 342.1 ms^{-1}
D) 348.1 ms^{-1}

Q.25 The value of "γ" for monochromatic gas is:
A) 1.67
B) 1.40
C) 1.29
D) 1.45

Q.26 According to Newton's formula of sound in air it STP is:
A) 332 ms^{-1}
B) 340 ms^{-1}
C) 350 ms^{-1}
D) 280 ms^{-1}

Q.27 Which of the following does not change when a wave changes its medium?
A) V
B) λ
C) f
D) All of them

Q.28 An engine is moving on a circular track with a constant sped. It is blowing whistle of frequency 500 Hz. The frequency received by an observer standing stationary at the center of the track is:
A) 500 Hz
B) More than 500 Hz
C) Less than 500 Hz
D) More or less than 500 Hz depending on the actual speed of the engine

Q.29 The first resonance length in a closed organ pipe is 50cm. Then the second resonance length will be:
A) 50cm
B) 100 cm
C) 150cm
D) 200 cm

Q.30 The number of nodes between two consecutive antinodes is:
A) Zero
B) 3
C) 2
D) 1

Q.31 A stretched string of length 2m vibrates in 4 segments. The distance between consecutive nodes is:
A) 0.5 m
B) 0.25 m
C) 1.0 m
D) 0.75 m

Q.32 In stationary waves:
A) Strain is maximum at antinodes
B) Strain is minimum at nodes
C) Strain is maximum at nodes
D) Amplitude is same at all points

Q.33 Two interfering waves have intensities in the ratio 9 : 1 then the ratio of maximum to minimum amplitude is:
A) 10 : 8
B) 4 : 2
C) 100 : 64
D) 16 : 4

Q.34 A man standing between two cliffs claps his hands and starts hearing a series of echoes at intervals of one second. If speed of sound in air is 340 m/s, then distance between the cliffs is:
A) 340 m
B) 680 m
C) 1020 m
D) 1360 m

Q.35 Phase difference between tow particles of a medium lying between two consecutive nodes in:
A) 0
B) $\pi / 4$
C) $\pi / 2$
D) π

Q.36 An air column in a pipe, which is closed at one end, will be in resonance with a vibrating tuning fork of frequency 256 Hz, if the length of the column in cm is (velocity of sound in air = 340ms^{-1}) :

A) 21.25
B) 125
C) 62.50
D) 33.2

Q.37 When one end of organ pipe of length 2m is closed, then the wavelength of stationary wave of 3rd harmonic is

A) 8
C) 8/3
B) 4
D) 8/5

Q.38 A observer is moving away from stationary source with speed 'v' of speed of sound, its frequency changed will be

A) f
C) f/2
B) 2f
D) zero

Q.39 A pipe is open from both ends, if the fundamental frequency is 10hz, find the length of the pipe, where speed of sound is 450m/s

A) 45m
C) 11.25m
B) 2.25m
D) 100cm

Q.40 The frequency of the note produced by plucking a given string increases as:

A) The length of the string increases
B) The tension in the string increases
C) The tension in the string decreases
D) The mass per unit length of the string increases

Q.41 An artificial satellite is receding from the Earth at a speed of 0.2 C where C is speed of light. It emits signals of frequency 4 x 10^7 Hz. The apparent frequency received is:

A) 3.3 x 10^7 Hz
B) 4 x 10^7 Hz
C) 3 x 10^8 Hz
D) None of these

Q.42 The ratio of the fundamental frequency of an open end and closed end organ pipes of the same length is:

A) 2: 1
B) 1: 2
C) 1: 1
D) 4: 1

Q.43 When two coherent waves interfere there is:

A) Loss of energy
C) redistribution of energy
B) Gain in energy
D) none of these

Q.44 Which of them is the result of collective motion of the particles of medium?

A) Mechanical wave
C) matter waves
B) Electromagnetic waves
D) none of these

Q.45 In standing wave if $\lambda = \frac{l}{2}$, in this case the number of loops are:

A) One
C) three
B) Two
D) four

Q.46 Transverse mechanical waves can be propagated:

A) Both is gas and metal
C) in metal but not in gas
B) In gas but not in metal
D) neither in gas not in metal

Q.47 A source is moving towards observer with velocity 100 m/s. The frequency of sound emitted by source is 500 Hz. The Doppler shift is:

A) 0.5 m
C) 5 m
B) 0.2 m
D) none of these

$\Delta\lambda = \frac{U}{f} = \frac{100}{500} = 0.2$ m

Q.48 In stretched string, the speed of wave is independent of:

A) Tension
C) thickness of given string
B) Length
D) the point from where the string is plucked

Q.49 Which of them is used for undersea communication?

A) Radio wave
C) gamma ray
B) X-rays
D) ultrasonic waves

Q.50 Frequency of infrasonic waves is:

A) From 20 Hz to 20 KHz
C) above from 20 KHz
B) Below from 20 Hz
D) none of these

THERMODYNAMICS (UNIT#05)

Q.1 When the state of gas change without change the temperature, the gas is said to undergo
A) Isothermal process
B) Adiabatic process
C) Isochoric process
D) Isobaric process

Q.2 We all know that Cp> Cv. What is also true for these molar specific heats (C_P, C_V)?
A) $\Delta Q_P > \Delta Q_V$
B) $\Delta U_P = \Delta U_V$
C) $\Delta T_P = \Delta T_V$
D) All of these

Q.3 During which process the change in the internal energy of the system is equal in magnitude to the work done by system :
A) Isobaric
B) isochoric
C) Adiabatic
D) isothermal process

Q.4 If $C_v = \frac{3}{2}R$, then the value of " γ ":
A) $\frac{5}{2}$
B) $\frac{3}{5}$
C) $\frac{5}{3}$
D) $\frac{2}{5}$

Q.5 The volume of the gas is reduced adiabatically to 1/4 of its volume at 27^0C. if γ =1.4, the new temperature will be:
A) $300\times(2)^{0.4}K$
B) $300\times(4)^{0.4}K$
C) $300\times(4)^{1.4}K$
D) $300\times(2)^{1.4}K$

$$TV^{\gamma-1} = Ccons\tan t \Rightarrow T_2 = \left(\frac{V_1}{V_2}\right)^{\gamma-1} T_1$$

$$T_2 = \left(\frac{V}{V/4}\right)^{1.4-1} 300 = 300\times(4)^{0.4}$$

Q.6 A system absorbs 80 J through heating while doing 100 J of external work. What is the change in the internal energy of the system?
A) -100 J
B) +80 J
B) -20 J
D) +180 J

$$\Delta U = Q - W$$
$$= 80 - 100$$
$$= -20J$$

Q.7 The adiabatic elasticity of hydrogen gas (γ = 1.4) at NTP is:
A) 1 x 10^5 Nm^{-2}
B) 1 x 10^{-8} Nm^{-2}
C) 1.4 Nm^{-2}
D) 1.4 x 10^5 Nm^{-2}

Q.8 In an adiabatic process, the is no:
A) Change in temperature
B) exchange of heat
C) Change in internal energy
D) work done

Q.9 Pressure temperature relationship for an ideal gas undergoing adiabatic change is (Letters have usual meanings):
A) PT^{γ} = constant
B) $PT^{-1+\gamma}$ = constant
C) $P^{\gamma-1}T^{\gamma}$ = constant
D) $P^{1-\gamma}T^{\gamma}$ = constant

Q.10 In an isothermal expansion:
A) Internal energy of the gas increases
B) Internal energy of the gas decreases
C) internal energy remains unchanged
D) Average kinetic energy of the gas molecules decreases

Q.11 For a certain gas the ratio of specific heats is given to be γ = 1.5. For this gas
A) C_v = 3R
B) C_p = 3R
C) C_v = 5R
D) C_p = 5R

Q.12 When a monoatomic ideal gas undergoes an isothermal change:
A) The number of degrees of freedom of the molecules changes
B) The temperature changes
C) There is no change of internal energy
D) There is no exchange of heat with the surroundings

Q.13 If a given mass of gas occupies a volume of 100 cc at one atmospheric pressure and Temperature of 100°C (373.15 K), what will be its volume at 4 atmospheric pressure, the temperature being the same?

A) 100 cc
B) 25 cc
C) 400 cc
D) 104 cc

Since T is constant:

$P_1V_1 = P_2V_2$

$V_2 = \frac{P_1 v_1}{P_2} = \frac{(1)(100)}{4} = 25$ cc.

Q.14 The ratio of rise in temperature of a gas when compressed adiabatically to that when compressed isothermally to the same extent:

A) Is less than 1
B) Is more than 1
C) depends on gas
D) is equal to 1

When gas is compressed adiabatically temperature rises T_a.
When gas is compressed isothermally temperature remains same $T_{iso.}$
Thus, $\frac{T_a}{T_{iso}} > 1$.

Q.15 Expansion during heating:

A) occurs at the same rate for all materials
B) occurs at different rates for different materials
C) increase the weight of a material
D) increase the density of material

Q.16 Two spheres of same size are made of the same material but one is hollow and the other is solid. They are heated to the same temperature. The:

A) both the spheres will expand equally
B) the hollow sphere will expand more than the solid one
C) the solid sphere will expand more than the hollow one
D) no conclusion can be drawn about their relative expansions unless the nature of the material is known

Q.17 A metal sheet with a circular hole is heated: The hole will:

A) contract
B) expand
C) remain unaffected
D) contract or expand depending on the value of the linear expansion coefficient

Q.18 During an adiabatic expansion the increase in volume is associated with:

A) decrease in pressure & decrease in temperature
B) increase in pressure & decrease in temperature
C) increase in pressure & increase in temperature
D) decrease in pressure & increase in temperature

Q.19 A liquid boils at a temperature at which its saturated vapor pressure becomes:

A) equal to the atmospheric pressure
B) twice the atmospheric pressure
C) half the atmospheric pressure
D) none of these

Q.20 The SI unit of Mechanical equivalent heat is:

A) joule-calories
B) joules/calorie
C) calorie-ergs
D) ergs/calorie

Q.21 During adiabatic compression of a gas its temperature:

A) remains constant
B) becomes zero
C) falls
D) rises

Q.22 A bicycle pump works in accordance with:

A) Newton's first law of motion
B) first law of thermodynamics
C) Pascal's law
D) Stoke's law

Q.23 Efficiency of Carnot engine becomes 100% if:

A) temperature of the source and the sink are equal
B) temperature of the source is 0 K (absolute zero)
C) temperature of the sink is 0 K (absolute zero)
D) none of these

Q.24 First law of thermodynamics when applied to an adiabatic process becomes:
A) W = ΔU
B) W = - ΔU
C) Q = ΔU
D) Q = W

Q.25 The ratio $\frac{C_p}{C_v} = \gamma$ for a diatomic gas like air is:
A) 1.67
B) 1.40
C) 1.30
D) 1.29

Q.26 The efficiency of diesel engine roughly ranges from:
A) 25% to 35%
B) 35% to 40%
C) 45% to 65%
D) 65% to 85%

Q.27 One Calorie is equal to:
A) 4.18 erg
B) 4.18 J
C) 41.8 J
D) 4.35 J

Q.28 At constant pressure, the graph between V (volume) and T (absolute temp.) is:
A) Hyperbola
B) Parabola
C) Straight line
D) Ellipse

Q.29 If the pressure is increased, the boiling point of the liquid:
A) Decrease
B) Remains constant
C) Increase
D) Decrease first and then increase

Q.30 Food is cooked quicker in a pressure cooker because:
A) Cooker is made of metal
B) It is a good conductor of heat
C) No heat is lost
D) High steam pressure raises the boiling point of liquid

Q.31 The temperature of source and sink of a Carnot engine are 327 ºC and 27 ºC, respectively. The efficiency of the engine is:
A) 1 – (27/327)
B) 27/327
C) 0.5
D) 0.7

$\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$

Q.32 In outer space there are 10 molecules per cm^3 on an average and the temperature there is 3 K. The average pressure of this light gas is:
A) 10^5 Nm^{-2}
B) 5×10^{-14} Nm^{-2}
C) 0.4×10^{-16} Nm^{-2}
D) 4.14×10^{-16} Nm^{-2}

$P = \frac{nKT}{V} = \frac{(10)(1.38 \times 10^{-23})(3)}{10^{-6}} = 4.14 \times 10^{-16}$ Nm^{-2}

Q.33 For a diatomic gas $C_v = 5R/2$, therefor gamma 'γ' for this gas is:
A) 5/7
B) 4/35
C) 7/5
D) 35/4

Q.34 At constant volume temperature is increased. Then:
A) Collision on walls will be less
B) Number of collision per unit time will increase
C) Collisions will be in straight lines
D) Collisions will no change

ELECTRO STYATICS (UNIT#06)

Q.1 In Coulomb's Law, the constant of proportional k = 1/4π ∈ *o* has units
A) N
B) N - m^2
C) NM^2/C^2
D) NC^2/M^2

Q.2 The magnitude of 1/4π ∈ *o* is:
A) 9×10^9
B) 9×10^{-9}
C) 8.85×10^{-12}
D) 8.85×10^{12}

Q.3 When a solid body is negatively charged by friction, it means that the body has:
A) Acquired excess of electron
B) lost some protons
C) Acquired some electron and lost a lesser number of protons
D) Lost some positive ions

Q.4 A suitable unit for expressing electric field strength is :
A) V/C
B) A-m
C) C/m^2
D) N/C

Q.5 A force of 0.01 N is exerted on a charge of 1.2 x 10^{-5} at a certain point. The electric field at that point is:
A) 5.3 x 10^4 N/C B) 8.5 x 10^4 N/C
C) 5.3 x 10^2 N/C D) 8.5 x 10^2 N/C

Q.6 The electric intensity at a point 20 cm away from a charge of 2 x 10^{-5} C is:
A) 4.5 x 10^6 N/C B) 3.5 x 10^5 N/C
C) 3.5 x 10^6 N/C D) 4.5 x 10^5 N/C

Q.7 Two positive charges of magnitudes 2 and 3 coulombs are placed 10 cm from the middle point on the right insect of the line joining the two charges is:
A) 5 x 10^{11} V B) 4 x 10^9 V
C) 4 x 10^{11} V D) 5 x 10^9 V

Q.8 Two charges are placed at a certain distance apart in vacuum. If a slab is placed between them, the force between them:
A) Will increase B) will decrease
C) Will remain unchanged
D) May increase or decrease depending on the material of the slab

Q.9 Four electric charges A, B, C and D are arranged as shown. The electric force will be least between charges:

A) A and B B) A and D
C) B and D D) A and C

Q.10 A and B are two points in an electric field. If the work done in carrying 4.0 coulomb of electric charge from A to B is 16.0 joule,teh potential difference between A and B is:
A) Zero B) 2.0 V
C) 4.0 V D) 16.0 V

Q.11 Two particles having charges Q_1 and Q_2, when kept at a certain distance, exert a force F on each other. If the distance between the two particles is doubled, the force between the particles would be:
A) 2F B) 4F
C) 8F D) 16F

Q.12 A positively charged particle of mass m kg and charge Q coulomb travels from rest through a P.D of V volt. Its kinetic energy in joule is:
A) QV B) mQV
C) mQ/V D) m/QV

Q.13 How many electrons will have a charge of one coulomb?
A) 6.2 x 10^{18} B) 6.2 x 10^{19}
C) 5.2 x 10^{18} D) 5.2 x 10^{19}

Q.14 An electron volt is equal to:
A) 0.62 x 10^{13} J B) 1.6 x 10^{-13} J
C) 0.62 x 10^{19} J D) 1.6 x 10^{-19} J

Q.15 Two point charges +2coulomb and +6 coulomb repel each other with a force of 12 N. If a charge of -2 coulomb is given to each of these charges, the force will now be:
A) Zero B) 8 N (attractive)
C) 8 N (repulsive) D) none of the above

Q.16 A hollow sphere of copper is positively charged. Then the electric field inside the sphere is:
A) the same as the field at the surface B) greater than the field at the surface
C) less than he field at the surface, but not zero
D) zero

Q.17 A and B are two points in an electric field. If 8.0 joule of work is done is taking 2.0 coulomb of electric charge from A and B is:
A) zero B) 2.0 V
C) 4.0 V D) 16.0 V

Q.18 Electric lines of force about a negative point charge are:

A) Circular, anticlockwise
B) circular, clockwise
C) Radial, inwards
D) radial, outwards

Q.19 Two plates are 1 cm apart and the potential difference between them is 10V. The electric field between the plates is:

A) 10 N/C
B) 250 N/C
C) 500 N/C
D) 1000 N/C

Q.20 As the electric charge on the surface of a hollow meal sphere increases, the electric field intensity inside the sphere:

A) Decreases
B) increases
C) Remains the same
D) May increase or decrease depending on the radius of the sphere

Q.21 Two point charges +2 coulomb and +6 coulomb repel each other with a force of 12 N. If a charge of -4 coulomb is given o each of these charges, the force will be:

A) 4 N (repulsive)
B) 4 N (attractive)
C) 8 N (repulsive)
D) 8 N (attractive)

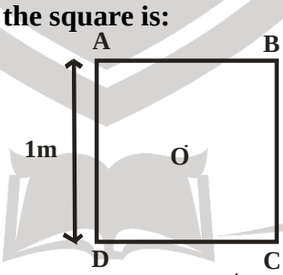
Q.22 The electrostatic force between two point charges q_1 and q_2 at separation r is given by $F\frac{Kq1q2}{r2}$

A) depends on the system of units only
B) depends on the medium between the charges only
C) depends on both the system of units and medium between the charges
D) is independent of both the system of units and the medium between the charges

Q.23 Three charge q, Q and 4q are placed in a straight line of length l at point's distance 0, l/2 and l respectively from one end. In Order to make the net force on q zero, the charge Q must be equal to:

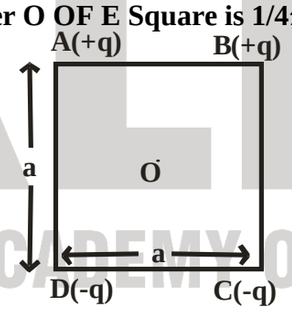
A) -q
B) -2q
C) $-\frac{q}{2}$
D) q

Q.24 Three charges, each of + 4μC, are placed at the corers B,C, D of a square ABCD of side 1m. The elecric field at the center O of the square is:



A) 7.2 x 10^4 N/C towards A
B) 7.2 x 10^4 N/C towards C
C) 3.6 x 10^4 N/C towards A
D) 3.6 x 10^4 N/C towards C

Q.25 Four charges +q, +q, - q and - q are placed respectively at the corners A, B, C and D of square of side a. The potential at the center O OF E Square is 1/4$\pi\epsilon_0$ times:



A) 4q
B) zero
C) $\frac{4\sqrt{2}q}{a}$
D) $\frac{4\sqrt{2}q}{a}$

Q.26 A capacitor having a capacity of 2.0 microfarad is charged up to 200v and its plates are joined to a wire. The heat produced in joule will be:

A) 4×10^{4} B) 4×10^{10}
C) 4×10^{-2} D) 4×10^{-2}

Q.27 Two capacitors of capacitance 0.3μF and 0.6μF area connected in series across a battery of 6V . The ratio of energies stored in them is :

A) 1/4 B) 1/2
C) 2 D) 4

Q.28 The S. I unit of permitivit of free space (ϵ_0) is

A) C/N-m B) N-m^2/C
C) C^2/(N-m)^2 D) C^2/N-m^2

Q.29 An electric charge is placed at the center of a cube of side a. The electric flux through one of is faces will be:

A) q / ϵ_0 B) q / ϵ_0 a^2
C) q / 4π ϵ_0 a^2 D) q / ϵ_0

Q.30 Two point charges placed at a distance r in air exert a force F on each other. The value of distance R at which they experience the same force when placed in a medium of dielectric constant k is:

A) rk B) $\frac{r}{k}$
C) r/$\sqrt{k}$ D) r$\sqrt{k}$

Q.31 A charged conductor has charge on its:

A) outside surface B) surrounding
C) middle point D) inner surface

Q.32 In comparison with the electrostatic force between two electrons, the electrostatice force between two protons is:

A) zero B) smallar
C) same D) greater

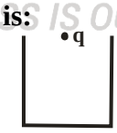
Q.33 No current flows between two charged bodies when connected if they have same:

A) capacity B) charge
C) potential D) none of the above

Q.34 The force between two charged particles is given by F= $\frac{1}{4\pi\varepsilon_0}\frac{Q_1Q_2}{r^2}$, where the symbols have their usual meaning. The dimensions of ϵ_0 free space in MKSQ system are:

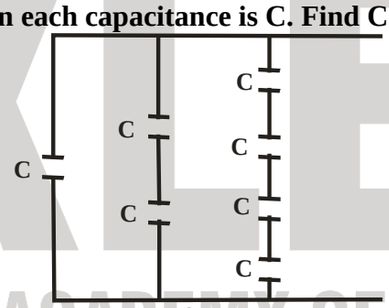
A) M^{-1}L^{2}T^{2}Q^4 B) ML^{-3}T^{4}Q^3
C) M^{-1}L^{-3}T^{4}Q^2 D) M^{-1}L^{-3}T^{2}Q^2

Q.35 A Charge q is placed at eh center of the open end of a cylindrical vessel. The flux of the electric field through the surface of the vessel is:



A) zero B) q / ϵ_0
C) q / 2 ϵ_0 D) 2q / ϵ_0

Q.36 In the capacitor network shown each capacitance is C. Find C_{eq}:



A) $\frac{7C}{4}$ B) $\frac{C}{2}$
C) $\frac{3C}{2}$ D) 2C

Q.37 A 100μF capacitor is charged to 200 volt. It is discharged through a 2 ohm resistance. The amount of heat generated will be:

A) 0.4 J B) 0.2 J
C) 2 J D) 4 J

Q.38 Farad is not equivalent to:
A) CV^2 B) J/V^2
C) Q^2/J D) Q/V

Q.39 The distance between the plates of a parallel plate air condenser is d. If the distance between the plates is reduced by half then the new capacitance will become:
A) Doubled B) half
C) one fourth D) remain unchanged

Q.40 The number of electrons in one coulomb charge is equal to:
A) 6.2×10^{18} B) 1.6×10^{-19}
C) Zero D) 6.02×10^{21}

Q.41 The value of Coulomb's constant K is:
A) 9×10^{-9} Nm^2C^{-2} B) 9×10^{9} N^2C^{-2}
C) 8.85×10^{-12} Nm^2C^{-2} D) 8.85×10^{12} Nm^2C^{-2}

Q.42 If the distance between the two charged bodies is halved, the force between hem becomes :
A) Double B) Half
C) Four Times D) One Half

Q.43 The Value of ϵ_o permittivity for free space is:
A) 1 B) 8.85×10^{-12} C^2 $(N^{-1}m^{-2})$
C) 9×10^{7} Nm^2C^2 D) 10^{-6} C

Q.44 Two chares are placed at a certain distance. If the magnitude of each charge is doubled, the force will become:
A) $1/4^{th}$ of its original value B) $1/8^{th}$ of its original value
C) 16 times of its original value D) 4 times of its original value

Q.45 The electric force of repulsion between two electrons at a distance of 1 m is:
A) 1.8 N B) 1.5×10^{-9} N
C) 2.30×10^{-28} N D) 2.30×10^{-27} N

Q.46 The unit of electric field strength (E) is:
A) Newton B) Coulomb
C) Joule/Coulomb D) Newton/coulomb

Q.47 Electric field intensity is a vector quantity and its direction is:
A) perpendicular to te direction of the field B) Opposite to the direction of force
C) Along the direction of force D) At a certain angle

Q.48 The electric intensity at infinite distance from the point charge is:
A) zero B) infinite
C) 1 volt m^{-1} D) positive

Q.49 The electric lines of force:
A) Physically exist around the charges B) Physically exist near the charges
C) Physically exist everywhere D) Imaginary

Q.50 According to the Statement of Gauss's Law, the flux through any closed surface:
A) $\emptyset_e = \varepsilon_o Q$ B) $\emptyset_e = Q / \varepsilon_o$
C) $\emptyset_e = \varepsilon_{o/Q}$ D) $\emptyset_e = Q^2 / \varepsilon_o$

Q.51 If a closed surface contains two equal and opposite carges, the net electric flux from the surface will be:
A) Positive B) Negative
C) Positive as well as negative D) zero

Q.52 The magnitude of Electric Intensity (E) at a point near an infinite sheet of positive charge is:
A) $E = \sigma / \varepsilon_o$ B) $E = \sigma / 2\varepsilon_o$
C) $E = \emptyset_e / A$ D) $E = \varepsilon_o / \sigma$

Q.53 The work done in moving a unit positive charge from one point to another against the electric field is a measure:
A) Intensity of electric field B) Resistance between two points
C) Capacitance D) Potential difference between two points

Q.54 If an electron isaccelerated through a potential difference V, it will acquire energy:
A) Ve B) V/2
C) E/V D) Ve^2

Q.55 The potential at a point situated at a distance of 50cm from a carge of 5μC is:
A) 9×10^{-4} volts B) 9×10^{-2} volts
C) 9×10^{4} volts D) 9×10^{2} volts

Q.56 A 50μF capacitor has a potential difference of 8V across it. The charge on the capacitor is:
A) 4×10^{-4} C
B) 4×10^{-3} C
C) 6.25×10^{-6} C
D) 6.25×10^{-5} C

Q.57 The value of permitivity in free space is:
A) 8.85×10^{12} $N^{-1}m^{-2}C^{-2}$
B) 8.85×10^{-10} $N^{-1}m^{-2}C^{-2}$
C) 8.85×10^{10} $N^{-1}m^{-2}C^{-2}$
D) 8.85×10^{-12} $N^{-1}m^{-2}C^{-2}$

Q.58 The electric field intensity at a point due to a point charge:
A) Falls of inversely as the distance
B) Falls of inversely as the square of distance
C) Remains unchanged with distance
D) Increases directly as square of distance

Q.59 If F represents the force between two point charges in free space (air), then the force between te same two charges in a medium of relative permitivity ϵ_r will be:
A) F' = F
B) F' = ϵ_r / F
C) F' = ϵ_r F
D) F' = F / ϵ_r

Q.60 The ration of the gravitational force F_g to the electrostatic force F_e between two electrons the same distance apart is approximately:
A) 9.8
B) 24×10^{19}
C) 24×10^{12}
D) 24×10^{-44}

Q.61 The force between two charges each of charge 1 μC placed 1m apart in air will be:
A) 9×10^{9} N
B) 9×10^{-12} N
C) zero
D) 9×10^{-3} N

Q.62 One coulomb of charge is created by:
A)10 electrons
B) 1.6×10^{-19} electrons
C) 6.25×10^{18} electrons
D) 6.25×10^{21} electrons

Q.63 1 kg mass of a subsace equivalent to the mass of:
A) 1.1×10^{30} electrons
B) 1.1×10^{19} electrons
C) 1.6×10^{31} electrons
D) 9.1×10^{-19} electrons

Q.64 The SI unit of electric flux is:
A) Weber
B) Nm^2C^{-1}
C) NmC^{-1}
D) $Nm^{-2}C$

Q.65 When an electron is acceleraed trough a P.D of one volt, it wil acquire energy equal to:
A) On joule
B) One erg
C) One electron volt
D) None of these

Q.66 The energy stored in an electric field between the plates of a capacitor per unit volume is:
A) Energy density = $\frac{1}{2}\epsilon_o \epsilon_r E^2$
B) Energy density = $(1/2\ \epsilon_o)\ \epsilon_r E^2$
C) Energy density = $(1/2\ \epsilon_o \epsilon_r)E^2$
D) Energy density = $\frac{1}{2}\epsilon_o \epsilon_r E$

Q.67 If time constant in R-C circuite is small, the capacitor is charged or discharged :
A) Rapidly
B) Slowly
C) At constant Rate
D) Intermittently

Q.68 The Electric intensity netween two oppsitetly charged parallel plates is given relation:
A) $E = \frac{\sigma}{2\varepsilon_o}\hat{\imath}$
B) $E = \frac{1}{4\pi\varepsilon_o}\frac{q}{r^2}\hat{\imath}$
C) $E = \frac{\sigma}{e_0}$
D) None of these

Q.69 One joule is equal to:
A) 1.6×10^{-19} V
B) 1.6×10^{19} V
C) 6.65×10^{-18} eV
D) 6.65×10^{18} eV

Q.70 Value of dielectric constant for air or vacuum is :
A) Less than one
B) Greater than one
C) one
D) zero

Q.71 The electric flux through any closed surface depends upon:
A) The charge only
B) The medium only
C) The medium and charge enclosed by the closed surface
D) The shape of the surface

ANSWERS

1.	(C)	**21.**	(B)	**41.**	(D)	**61.**	(D)
2.	(A)	**22.**	(C)	**42.**	(C)	**62.**	(C)
3.	(A)	**23.**	(A)	**43.**	(B)	**63.**	(A)
4.	(D)	**24.**	(A)	**44.**	(D)	**64.**	(B)
5.	(D)	**25.**	(B)	**45.**	(C)	**65.**	(C)
6.	(A)	**26.**	(C)	**46.**	(D)	**66.**	(A)
7.	(C)	**27.**	(C)	**47.**	(C)	**67.**	(A)
8.	(B)	**28.**	(D)	**48.**	(A)	**68.**	(C)
9.	(D)	**29.**	(A)	**49.**	(D)	**69.**	(D)
10.	(C)	**30.**	(C)	**50.**	(B)	**70.**	(C)
11.	(D)	**31.**	(A)	**51.**	(D)	**71.**	(C)
12.	(A)	**32.**	(C)	**52.**	(B)	**72.**	
13.	(A)	**33.**	(C)	**53.**	(D)	**73.**	
14.	(D)	**34.**	(D)	**54.**	(A)	**74.**	
15.	(D)	**35.**	(C)	**55.**	(C)	**75.**	
16.	(D)	**36.**	(A)	**56.**	(A)	**76.**	
17.	(C)	**37.**	(C)	**57.**	(D)	**77.**	
18.	(C)	**38.**	(A)	**58.**	(B)	**78.**	
19.	(D)	**39.**	(A)	**59.**	(D)	**79.**	
20.	(C)	**40.**	(A)	**60.**	(D)	**80.**	

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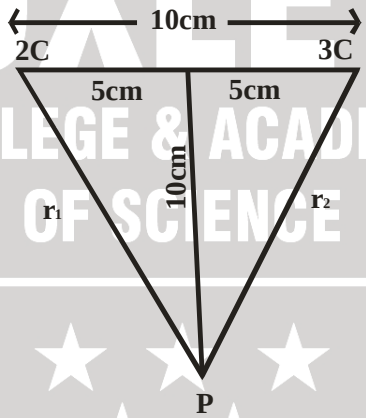
SOLUTION / HINTS

5. The Electric Field at a point due to a point charge is given by:

$$E = \frac{F}{q}$$

7.



The electric potential at P is given by:

$$V = V_1 + V_2 \ldots\ldots\ldots\ldots\ldots\ldots\text{(i)}$$

Here $V_1 = \frac{1}{4\pi\epsilon_o}\left(\frac{2}{r_1}\right)$

$V_2 = \frac{1}{4\pi\epsilon_o}\left(\frac{3}{r_2}\right)$

Here $V = \frac{1}{4\pi\epsilon_o}\left(\frac{2}{0.111} + \frac{3}{0.111}\right)$

$V = (9 \times 10^9)(18.02 + 27.02)$

$V = 4 \times 10^{11}$ V

Here $r_1 = r_2 = \sqrt{5^2 + 10^2}$

$= \sqrt{125}$ cm

$= 11.18$ cm

$= 0.111$ cm

13. $1\ e = 1.6 \times 10^{19}$ C

$1C = \frac{1}{1.6 \times 10^{19}} = 6.02 \times 10^{18}$ electron

21. $F = \frac{K(2)(6)}{r^2} = 12$

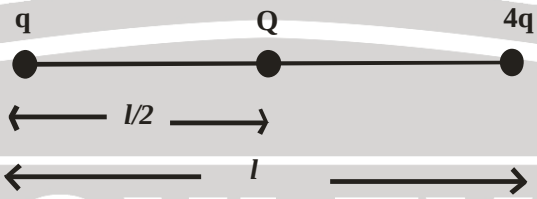
$\frac{K}{r^2} = 1$

When -4 is given to each charge, they become -2C and +2C. The new force is:

$F' = \frac{K(2)(2)}{r^2}$ (attractive)

$= 4$ N (attractive)

23.



The net force on charge q due to charge 4q and Q is given by:

$$F = K\left[\frac{qQ}{(\frac{l}{2})^2} + \frac{4q^2}{l^2}\right]$$

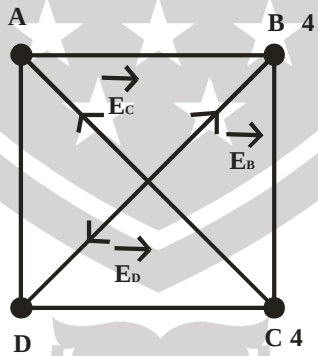
$$F = K\left[\frac{4qQ}{l^2} + \frac{4q^2}{l^2}\right] \text{ .....(i)}$$

Let F = 0, therefore, from equation (i):

$$\frac{4qQ}{l^2} = -\frac{4q^2}{l^2}$$

$\Rightarrow$ Q = - q

24.



$\overrightarrow{E_B}$ and $\overrightarrow{E_D}$ cancel out. Therefore the net electric field is E_c , which is towards A and has the magnitude.

$$\frac{9 \times 10^9 \times 4 \times 10^{-6}}{1/\sqrt{2}} = 7.2 \times 10^4 \text{ N}$$

25. The potential at the center 'O' of the square is:

$$V = \frac{q}{4\pi\varepsilon_o}\left[\frac{1}{OA} + \frac{1}{OB} + \frac{1}{OC} + \frac{1}{OD}\right]$$

$$V = 0$$

(∵ OA = OB = OC = OD)

26. The heat produced in joule is calculated as:

$$U = \frac{1}{2}CV^2$$

$$= \frac{1}{2}(2 \times 10^{-6})(200)^2$$

$$= 0.04 = 4 \times 10^{-2}$$

27. Since the capacitors are in series, the charge o bot is the same ad is equal to the charge o the equivalent capacitor. Therefore,

$E_1 = \frac{1}{2}CV^2 = \frac{1}{2}\frac{Q^2}{C_1}$

And $E_2 = \frac{1}{2}\frac{Q^2}{C_2}$

$\frac{E_1}{E_2} = \frac{2C_1}{\frac{1}{2}\frac{Q^2}{C_2}} = \frac{C_2}{C_1} = \frac{0.6\ \mu F}{0.3\ \mu T} = 2$

29. According to Gauss's law, the electric flux through the cube = q/ϵ_o. Since there are six faces the flux through one face = $q/6\epsilon_o$.

30. $F = \frac{1}{4\pi\varepsilon_o}\frac{q_1q_2}{r^2} = \frac{1}{4\pi\varepsilon_o K}\frac{q_1q_2}{R^2}$

$R = \frac{r}{\sqrt{K}}$

36. From the given circuit:

$C_{eff} = C + \frac{C}{2} + \frac{C}{4} = \frac{7C}{4}$

Note: Capacitors in series:

If $C_1, C_2, \ldots\ldots, C_n$ are in series then:

$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \ldots\ldots\ldots \frac{1}{C_n}$

Capacitors in parallel:

If $C_1, C_2, \ldots\ldots, C_n$ are in parallel then:

$C_{eq} = C_1 + C_2 + \ldots\ldots\ldots C_n$

37. $H = \frac{1}{2}CV^2 = \frac{1}{2} \times 10^{-4} (200)^2 = 2$ J

CURRENT ELECTRICITY (UNIT#07)

Q.1 A steady current is flowing in a conductor of non-uniform cross-section. The charge passing through any cross-section per unit time is:

A) directly proportional to the area of cross-section
B) inversely proportional to the area of cross-section
C) proportional to square of the area of cross-section
D) independent of the area of cross-section

Q.2 A copper wire of length 2m and area of cross-section 1.7×10^{-6} m^2 has a resistance of 2×10^{-2} ohm. The resistivity of copper is:

A) 1.7×10^{-8} ohm-meter
B) 1.9×10^{-8} ohm-meter
C) 2.1×10^{-7} ohm-meter
D) 2.3×10^{-7} ohm-meter

Q.3 If the length of the wire is doubled and its cross- section is also doubled, then its resistance will:

A) become four times
B) become one-fourth
C) become two times
D) remain unchanged

Q.4 A man has five resistors each of value 1/5 ohm. The minimum resistance he can obtain by combining them is:

A) 1/50 ohm
B) 1/25 ohm
C) 1/10 ohm
D) none of these

Q.5 Combining the five resistances each of value $\frac{1}{5}$ ohm, the maximum resistance that can be obtained is:

A) 1 ohm B) 1/2 ohm
C) 2/5 ohm D) none of these

Q.6 A resistance of 6 ohm is connected in series with another resistance of 4 ohm across a battery of 20 V. The P.D. across the 6 ohm resistor is:

A) 3V B) 6V
C) 6V D) 12V

Q.7 Three resistors of 2 ohm, 3 ohm and 5 ohm are connected in parallel across a battery of 10 V and of negligible internal resistance. The P.D. across the 3 ohm resistor is:

A) 2V B) 3V
C) 9V D) 10V

Q.8 The resistance of a wire is R. It is stretched uniformly so that its length is doubled. The resistance now becomes:

A) 2R B) R/2
C) 4R D) R/4

Q.9 A wire has resistance 12Ω . It is bent in the form of a circle. The effective resistance between two points across a diameter is:

A) 3 Ω B) 6 Ω
C) 12 Ω D) 24 Ω

Q.10 The wires of the same material have length 6 cm and 10 cm and radii 0.5 mm and 1.5 mm respectively. They are connected in series across a battery of 32 V. The P.D. across the shorter wire is:

A) 5V B) 13.5V
C) 10V D) 27V

Q.11 The specific resistance of a wire :

A) varies with its length B) varies with its cross-section
C) varies with its mass
D) does not depend on its length, cross-section and mass

Q.12 A current of 4.8 A is flowing in a conductor. The number of electrons passig through any cross-section per second is:

A) 3×10^{19} B) 3×10^{20}
C) 7.68×10^{19} D) 7.68×10^{20}

Q.13 The equivalent resistance of two wires in parallel is 6/5 Ω . If the resistance of one of the wire is 2 Ω , that of the other is:

A) 3/5 B) 2
C) 5/3 D) 3

Q.14 The resistance of the wire is R. It is cut into four equal parts and all the parts are bundled together side by side. The resistance of the bundle is:

A) R/16 B) R/8
C) R/4 D) R

Q.15 Three resistances each of 4 Ω , are connected to form a triangle. The resistance between any two terminals is:

A) 2 Ω B) 8/3 Ω

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C) 6 Ω D) 12 Ω

Q.16 A cell of negligible resistance and emf 2V is connected across a series combination of 2, 3 and 5 ohms. T P.D. across the 3 Ω resistor is:

A) 0.6 V B) 2/3 V
C) 1/3 V D) 4/3 V

Q.17 A wire of length 5 m and radius 1 mm has a resistance of 1 Ω. What length of a wire of same maerial but of radius 2 mm will also have a resistance of 1 Ω ?

A) 1.25m B) 2.5m
C) 10m D) 20m

Q.18 A current of 2.0 A passes through a cell of emf 1.5 V, having internal resistance 0.15 Ω. The potential difference, in volts, across the ends of the cell will be:

A) 1.35 B) 1.50
C) 1.00 D) 1.20

Q.19 The resistances R_o and R_t of a metallic wire at temperatures 0°C and r°C are related as (α is the temperature coefficient of resistance):

A) $R_t = R_o (1 + \alpha t)$ B) $R_t = R_o (1 - \alpha t)$
C) $R_t = R_o^2 (1 + \alpha t)$ D) $Rt = R_o^2 (1 - \alpha t)$

Q.20 The reciprocal of specific resistance is:

A) conductive resistance B) specific conductance
C) conductive reactance D) plate resistance

Q.21 A flow of 10^7 electron per second in a conduction wire constitutes a current of:

A) 1.6×10^{-26} A B) 1.6×10^{12} A
C) 1.6×10^{-12} A D) 1.6×10^{26} A

Q.22 Two resistance, 4 Ω and 6 Ω, are is series and a 10 Ω resistor is in parallel to combination. The resultant resistance is:

A) 5 Ω B) 8 Ω
C) 12 Ω D) 20 Ω

Q.23 Which of the following statements is not true?

A) Conductance is the reciprocal of resistance and is measured in siemens
B) Ohm's law is not applicable at a very low and a very high temperature
C) Ohm's law is applicable to semiconductor
D) Ohm's law is not applicable to electron tubes, discharge tubes and electrolytes

Q.24 Kirchhoff's first law, i.e., ∑ I = 0 at a junction, deals with conservation of:

A) charge B) energy
C) momentum D) angular momentum

Q.25 A carbon resistance has colour bands in order: yellow, brown, red. It resistance is:

A) 41 Ω B) 41×10^2 Ω
C) 41×10^3 Ω D) 4.2 Ω

Q.26 The resistivity of a wire is depends on its:

A) length B) area of cross-section
C) shape D) material

Q.27 The conductivity of a superconductor is:

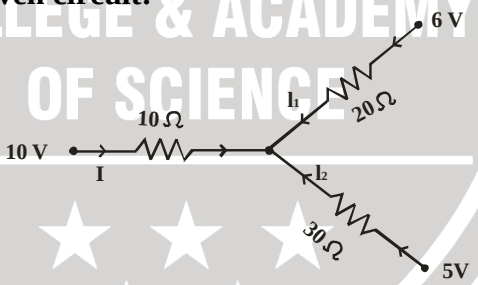
A) infinite B) very large
C) very small D) zero

Q.28 Kirchhoff's voltage law is based on the law of conservation of:
A) Momentum B) Charge
C) Mass D) Energy

Q.29 Electromotive force is most closely related to:
A) Electric field B) Magnetic field
C) Potential difference D) Mechanical force

Q.30 Potentiometer measures potential more accurately because:
A) it measures potential in the open circuit
B) it uses sensitive galvanometer for null detection
C) it uses high resistance potentiometer wire
D) it measures potential in the closed circuit

Q.31 Find current I in the given circuit:



A) 0.2 A B) 0
C) 9.1 A D) none of these

Q.32 Drift velocity of electrons in a conductor is of the order of:
A) 10^6 ms^{-1} B) 10^4 ms^{-1}
C) 10^{-3} ms^{-1} D) 10^{-9} ms^{-1}

Q.33 Ampere-second stands for the unit of :
A) emf B) energy
C) charge D) power

Q.34 Two unequal resistances are connected parallel across a battery. Which of the following statement is true?
A) same current will flow through both resistances
B) current through smaller resistance is higher
C) current through larger resistance is higher
D) current can be higher in any resistance depending on emf of the cell

Q.35 A 100 W bulb rated at 220 V is connected to a 110 V supply. The power consumed is nearly equal to:
A) 100 W B) 50 W
C) 25 W D) 12.5 W

Q.36 The fuse wire is a wire of:
A) low resistance and high melting point
B) high resistance and high melting point
C) high resistance and low melting point
D) low resistance and low melting point

Q.37 The heating element of and electric heater should be made of a material which should have:
A) high specific resistance and high melting point
B) low specific resistance and low melting point

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C) high specific resistance and low melting point
D) low specific resistance and high melting point

Q.38 You are given three bulbs of 25, 40 and 60 watt. Which of them has lowest resistance?
A) 25 W bulb
B) 40 W bulb
C) 60 W bulb
D) none of these

Q.39 The maximum current that can be allowed to pass through 100W / 250V lamp is:
A) 0.25 V
B) 0.40 A
C) 2.5 A
D) 1.00 A
P = VI

Q.40 If a charge 'Q' flows through any cross-section of the conductor in time 't' second, the current 'I' is given by:
A) I = Qt
B) I = Q/t
C) I = t/Q
D) $I = Q^2/t$

Q.41 One coulomb per second is equal to:
A) One Volt
B) One Ampere
C) One Watt
D) One Ohm

Q.42 The graphical representation of Ohm's law is:
A) Hyperbola
B) Ellipse
C) Parabola
D) Straight

Q.43 The resistance of a conductor of length L cross-section area A and resistivity ρ is given by:
A) $R = \rho A / L$
B) $R = \rho L / A$
C) $R = \rho / LA$
D) $R = A / \rho A$

Q.44 The resistance of a meter cube of the substance is called:
A) Conductivity
B) Permittivity
C) Resistivity
D) Susceptibility

Q.45 A wire of uniform area of cross-section 'A' length L and resistance R is cut into two equal parts. The resistivity of each part:
A) Is doubled
B) Remains same
C) Is halved
D) Is one-forth

Q.46 The fractional change in resistivity per unit original resistivity per Kelvin is known as:
A) Temperature co-efficient of resistance
B) Temperature co-efficient of conductance
C) Temperature co-efficient of conductivity
D) Temperature co-efficient of resistivity

Q.47 If the resistors are connected in parallel, then:
A) The current through each is the same
B) The total resistance is the sum of individual resistors
C) The voltage across each is the same
D) The total resistance is the product of the individual resistors

Q.48 Three resistances 5000, 500 and 50 ohms are connected in series across 555 volts mains. The current flowing through them will be:
A) 1 A
B) 100 mA
C) 10 mA
D) 10 A

Q.49 Heat energy dissipated in a resistor R when connected to a battery of V volts and current I ampere flowing through it for time t is given by:

A) I^2R B) IRt
C) VIt D) V^2 / Rt

Q.50 The resistance of a 60 watt bulb in a 120 volt line is:

A) 20 ohms B) 0.5 ohms
C) 240 ohms D) 2.0 ohms

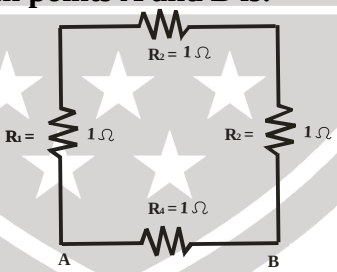
Q.51 The electromotive force of a battery or cell is the voltage between its terminals when:

A) The circuit is open B) The circuit is closed
C) Its internal resistance is minimum D) Its internal resistance is maximum

Q.52 The SI unit of electromotive force is:

A) Coulomb per second B) Joule per coulomb
C) Coulomb per volt D) Volt per ampere

Q.53 Four resistors each of value one ohm are connected as shown in fig below. The equivalent resistance between points A and B is:



A) zero B) $\frac{3}{4}\Omega$
C) $\frac{4}{3}\Omega$ D) 4 Ω

Q.54 Drift velocity of electrons in a conductor is of the order of:

A) 10^{-3} ms^{-1} B) 10^{3} ms^{-1}
C) 10^{-4} ms^{-1} D) 10^{4} ms^{-1}

Q.55 A gold band shows a tolerance of:

A) ± 10% B) ± 7%
C) ± 5% D) ± 20%

Q.56 Kirchhoff's first rule is the manifestation of law of conservation of:

A) Mass B) Energy
C) Momentum D) Charge

Q.57 Which one of the following statements is correct:

A) The current flowing from the negative towards positive terminal of the battery is called electronic current.
B) Electronic current is constituted by the flow of electrons
C) Electronic current always flows from a lower to higher potential
D) All the above

Q.58 The temperature coefficient of resistance is expressed as:

A) $\alpha = \frac{R_t - R_0}{R_o t}$ B) $\alpha = \frac{R_t - R_0}{t}$
C) $\alpha = \frac{R_t - R_0}{R_1 t}$ D) $\alpha = \frac{R_0 t}{R_t - R_0}$

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Q.59 The resistance of a conductor one meter in length and one square meter in area of cross-section is called is:
A) Reactance
B) Conductance
C) Resistivity
D) Conductivity

Q.60 If a wire of resistance R is stretched to double its length, its resistance becomes:
A) $\frac{R}{2}$
B) 4R
C) 2R
D) $\frac{R}{4}$

Q.61 N identical resistors each of resistance R are connected in parallel. The equivalent resistance of the combination will be:
A) NR
B) $\frac{R}{N}$
C) $\frac{N}{R}$
D) none of these

Q.62 Terminal P.D. of a battery of internal resistance r and emf E is:
A) $V_t = E + IR$
B) $V_t = E - IR$
C) $V_t = \frac{E-R}{R}$
D) $V_t = \frac{E-R}{t}$

Q.63 The amount of heat energy dissipated in a resistance is directly proportional to:
A) The square of the current only
B) The resistance of conductor only
C) The time of current flow only
D) The square of the current, resistance and time of current flow

Q.64 When the internal resistance of the source equals the load resistance, the maximum output power is given by:
A) $P_{out} = \frac{E^2}{4r}$
B) $P_{out} = E^2 \times r$
C) $P_{out} = 4E^2r$
D) $P_{out} = \frac{E^2}{r}$

Q.65 Kirchhoff's second rule (KVL) is related to:
A) IR drops
B) Battery emfs
C) Junction voltages
D) Both (a & b)

Q.66 Three resistors each of one ohm are connected to form triangle. The resistance between any two terminals is:
A) $\frac{2}{3}\Omega$
B) $\frac{3}{2}\Omega$
C) $3\ \Omega$
D) $1\ \Omega$

Q.67 The force per unit charge is known as:
A) Electric flux
B) Electric intensity
C) Electric potential
D) Electric current

Q.68 An ideal voltmeter should have resistance:
A) Infinity
B) zero
C) Some resistance
D) Between zero and infinity

Q.69 Resistance of a conductor depends upon:
A) The quantity of current passing through it
B) The voltage applied between its ends
C) Its dimensions, physical state and nature of material
D) All of the above

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Q.70 The electromotive force of a cell or battery is the voltage between its terminals, when:

A) It is in an open circuit
B) Its internal resistance is maximum
C) Its internal resistance is minimum
D) It gives current to same appliance

Q.71 Electric charge on an object is measured as 5 micro coulomb. How the value of this charge can be expressed in terms of base units:

A) 5×10^{6} ampere second.
B) 5×10^{-6} ampere second.
C) $5 \times 10^{+6}$ coulomb second.
D) 5×10^{0} coulomb second.

Q.72 If a resistor having resistance 'R' is cut into three equal parts, then equivalent of parallel combination is:

A) 0/R
B) 3/R
C) R/9
D) R/3

Q.73 Which combination of several identical resistors each of 2Ω gives rise to the resultant of 10/11Ω

A) 5 parallel, 2 series
B) 4 parallel, 3 series
C) 3 parallel, 4 series
D) 2 parallel, 5 series

Q.74 The resistance of piece of wire is 12Ω. It is bent to form an equilateral triangle. What is the equivalent resistance between any two corners of the triangle?

A) 1.3Ω
B) 2.0Ω
C) 4.0Ω
D) 2.7Ω

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ANSWERS

1.	(D)	**21.**	(C)	**41.**	(B)	**61.**	(B)
2.	(A)	**22.**	(A)	**42.**	(D)	**62.**	(B)
3.	(D)	**23.**	(C)	**43.**	(B)	**63.**	(D)
4.	(B)	**24.**	(A)	**44.**	(C)	**64.**	(A)
5.	(A)	**25.**	(B)	**45.**	(B)	**65.**	(D)
6.	(D)	**26.**	(D)	**46.**	(D)	**66.**	(A)
7.	(D)	**27.**	(A)	**47.**	(C)	**67.**	(B)
8.	(C)	**28.**	(D)	**48.**	(B)	**68.**	(A)
9.	(A)	**29.**	(C)	**49.**	(C)	**69.**	(C)
10.	(D)	**30.**	(A)	**50.**	(C)	**70.**	(A)
11.	(D)	**31.**	(A)	**51.**	(A)	**71.**	(B)
12.	(A)	**32.**	(C)	**52.**	(B)	**72.**	(C)
13.	(D)	**33.**	(C)	**53.**	(B)	**73.**	(D)
14.	(A)	**34.**	(B)	**54.**	(A)	**74.**	(B)
15.	(B)	**35.**	(C)	**55.**	(C)	**75.**	
16.	(A)	**36.**	(D)	**56.**	(D)	**76.**	
17.	(D)	**37.**	(A)	**57.**	(D)	**77.**	
18.	(D)	**38.**	(C)	**58.**	(A)	**78.**	
19.	(A)	**39.**	(B)	**59.**	(C)	**79.**	
20.	(B)	**40.**	(B)	**60.**	(B)	**80.**	

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Solutions / Hints

2. $\rho = \frac{RA}{L}$

4. The minimum resistance can be obtained if these resistors are connected in parallel:

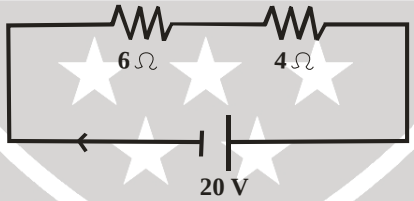
i.e. $\frac{1}{R} = \frac{1}{1/5} + \frac{1}{1/5} + \frac{1}{1/5} + \frac{1}{1/5} + \frac{1}{1/5}$

$\frac{1}{R} = 5 + 5 + 5 + 5 + 5$

$\frac{1}{R} = 25$

$R = \frac{1}{25}$

6.



$I = \frac{20\,V}{10\Omega}$

$I = 2A$

And P.D. across 6Ω is:

$V = IR$

$= (2A)(6\Omega)$

$V = 12\ V$

7. In parallel combination the P.D. across each resistor is same.

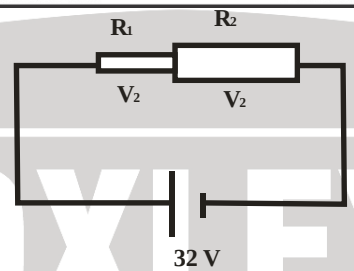
8. $R = \frac{\rho L}{A}$

When l is doubled, A becomes half. So,

$\acute{R} = \frac{\rho(2l)}{A/2} = 4(\frac{\rho l}{A}) = 4R$

9. It becomes two resistors, each of 6 Ω connected in parallel.

10. Let the length, radius and resistance of the shorter wire be l_1, r_1 and R_1, respectively, and the corresponding quantities for the longer wire be l_2, r_2 and R_2 respectively. Then



$R_1 = \frac{\rho l_1}{\pi r_1^2}$, $R_2 = \frac{\rho l_2}{\pi r_1^2}$

$\frac{R_1}{R_2} = \frac{l_1 r_2^2}{l_2 r_1^2} = \frac{3}{5} \times \left[\frac{3}{1}\right]^2 = \frac{27}{5}$

Let V_1 and V_2 be the P.D. across the shorter and longer wire, respectively. Then

$\frac{V_1}{V_2} = \frac{R_1}{R_2} = \frac{27}{5}$ ……(i)

Also $V_1 + V_2 = 32$……(ii)

From (i) and (ii):

$V_1 = 27$ V

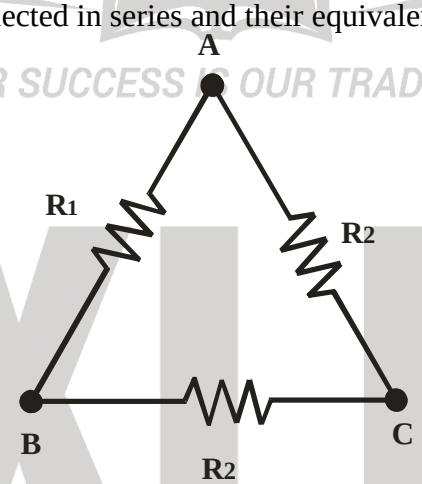
12. $I = \frac{Q}{t} = \frac{ne}{t}$

$\Rightarrow n = \frac{It}{e} = \frac{(4.8A)(1\ sec)}{1.6 \times 10^{-19} C}$

$n = 3 \times 10^{19}$

15. Let we find resistance between B and C.

Here R_1 and R_2 are connected in series and their equivalent resistance is parallel to R_3.



Therefore,

$\frac{1}{R_{BC}} = \frac{1}{R_3} + \frac{1}{R'}$ (Here $R' = R_1 + R_2$)

$\frac{1}{R_{BC}} = \frac{1}{4} + \frac{1}{8}$

$\frac{1}{R_{BC}} = \frac{3}{8}$

$R_{BC} = \frac{8}{3}\ \Omega$

17. $R_1 = \frac{\rho l}{\pi r_1^2}$ ……(i)

$R_2 = \frac{\rho l}{\pi r_2^2}$ ……..…..(ii)

Here

$l_1 = 5m$

$r_1 = 1mm$

$l_2 = ?$

$r_2 = ?$

Using $R_1 = R_2$

$l_2 = 20m$

18. $V = E - Ir$

21. $I = \frac{Q}{t}$

$I = \frac{10^7 e}{t} = \frac{10^7 (1.6 \times 10^{-19} C)}{1\ sec}$

$I = 1.6 \times 10^{-12}\ A$

31. Apply Kirchhoff's Rule:

$I = I_1 + I_2$

$\frac{10-V}{10} = \frac{V-6}{20} + \frac{V-5}{30}$

Or

$\frac{5.5V}{30} = 1 + \frac{3}{10} + \frac{1}{6}$

$\frac{5.5V}{30} = \frac{30+9+5}{30}$

Or $V = \frac{44}{5.5} = 8\ V$

$I = 0.2\ V$

35. $R = (220)^2 = 484\ \Omega$

Power Consumed $= \frac{V^2}{R} = \frac{110 \times 110}{484} = 25W$

36. Low resistance consumes less power and low melting point will protect the device when a large current enters.

38. $P = \frac{V^2}{R}$

ELECTROMAGNETISM (UNIT#08)

Q.1 A unit of magnetic induction is:
A) weber/meter^2 B) weber/ampere-meter
C) weber/meter D) weber

Q.2 The permeability of free space, in Wb/Am, is:
A) $\frac{1}{4\pi \times 10^{-7}}$ B) $\frac{1}{4\pi \times 10^{7}}$
C) $4\pi \times 10^{-7}$ D) $4\pi \times 10^{7}$

Q.3 One Wb/m^2 is equal to:
A) 10^4 gauss B) 10^2 gauss
C) 10^{-2} gauss D) 10^{-4} gauss

Q.4 An electric charge q moves with a constant velocity v parallel to the lines of force of a uniform magnetic field B. The force experienced by the charge is:
A) Bqv B) B/qv
C) Bq/v D) zero

Q.5 A proton enters a magnetic field of flux density 1.5 Wb/m^2 with a speed of 2 x 10^7 m/s at angle of 30° with the field. The force on the proton will be:
A) 0.24 x 10^{-12} N B) 2.4 x 10^{-12} N
C) 24 x 10^{-12} N D) 0.024 x 10^{-12} N

Q.6 Which of the following does not affect the motion of a moving electron?
A) Electric field applied in the direction of motion
B) Magnetic field applied I the direction of motion
C) Electric field applied perpendicular to the direction of motion
D) Magnetic field applied perpendicular to the direction of motion

Q.7 An electron is moving in a circle of radius r in a uniform magnetic field B. Suddenly he field is reduced to B/2. The radius of the circle now becomes:
A) r/2 B) r/4
C) 2r D) 4r

Q.8 A charged particle moving with a velocity $\vec{v}$ enters a uniform magnetic field $\vec{B}$. The particle experiences the largest deflecting force when the angle between $\vec{v}$ and $\vec{B}$ is:
A) 0° B) 45°
C) 90° D) 180°

Q.9 A moving charge produces:
A) an electric field only B) a magnetic field only
C) both electric and magnetic fields D) neither electric or a magnetic field

Q.10 The strength of e magnetic field due to a long straight conductor is:
A) the same everywhere around the conductor
B) inversely proportional to the distance from the conductor
C) inversely proportional to the square of the distance from the conductor
D) inversely proportional to the cube of the distance from the conductor

Q.11 Two free parallel wires carrying currents in opposite directions:
A) do not effect each other
B) attract each other
C) repel each other
D) get rotated to be perpendicular to each other

Q.12 In a moving coil galvanometer the current *i* is related to the deflection θ as:
A) $i \propto \theta$
B) $i \propto tan\ \theta$
C) $i \propto \theta^2$
D) $i \propto \sqrt{\theta}$

Q.13 A long solenoid as 20 turns/cm. The current necessary to produce a magnetic field of 20 millitesla inside the solenoid is approximately:
A) 1 A
B) 2 A
C) 4 A
D) 8 A

Q.14 An electron is projected in a magnetic field along the lines of force.
A) There will be no effect on the motion of electron
B) The electron speed will increase but the direction will not change
C) The electron speed will decrease but the direction will not change
D) The electron speed will remain the same but it will move in a circle

Q.15 A strong magnetic field is applied on a stationary electron. Then the electron:
A) moves in the direction of field
B) moves opposite to the field
C) starts spinning
D) remains stationary

Q.16 Magnetic effect of current was discovered by:
A) Faraday
B) Oersted
C) Kirchhoff
D) Joule

Q.17 The unit of magnetic flux is:
A) weber
B) weber/m^2
C) henry
D) ampere/m

Q.18 A moving coil galvanometer as a coil of effective area A and number of turns N. The magnetic field B is radial. If a current I is passed through the coil, the torque acting on the coil is;
A) NA^2B^2I
B) $NABI^2$
C) N^2ABI
D) NABI

Q.19 A circular loop of area 0.01 m^2 and carrying a current of 10A is parallel to a magnetic field of intensity 0.1 T. The torque acting on the loop, I Nm, is:
A) zero
B) 0.8
C) 0.001
D) 0.01

Q.20 Lorentz force is given by:
A) q(E + v x B)
B) q(E - v x B)
C) q(E + v . B)
D) q(E x B x v)

Q.21 A magnetic field:
A) always exerts a force on a charged particle
B) never exerts a force on a charged particle
C) exerts a force on a charged particle if it is moving along the magnetic lines of force

D) exerts a force on a charged particle if it is moving along the magnetic lines of force

Q.22 A straight wire of diameter 0.5 mm carrying a current of 1A is replaced by another wire of diameter 1 mm carrying same current. The strength of the magnetic field far away is:

A) twice the earlier value
B) half of the earlier value
C) quarter of the earlier
D) not changed

Q.23 The magnetic field lines in the middle of a solenoid are:

A) circles
B) parallel to the axis
C) spirals
D) perpendicular to the axis

Q.24 If a current is passed in a spring, it:

A) gets compressed
B) gets expanded
C) oscillates
D) remains unchanged

Q.25 A straight wire of length 0.5 m and carrying a current of 1.2 A is placed in a uniform magnetic field of induction 2T. The magnetic field is perpendicular to the length of wire. The force on the wire is:

A) 2.4 N
B) 1.2 N
C) 3.0 N
D) 2.0 N

Q.26 The expression for magnetic induction inside a solenoid of length L, carrying a current I and having N turns is:

A) $\frac{\mu_o}{4\pi}\frac{N}{L}I$
B) $\mu_o NLI$
C) $\frac{\mu_o}{4\pi}NLI$
D) $\mu_o \frac{N}{L}I$

Q.27 A charged particle is moving with velocity v in a magnetic field of induction B. The force on the particle will be maximum when:

A) v and B are in the same direction
B) v and B are in the opposite directions
C) v and B are parallel
D) v and B are at an angle of 45°

Q.28 A current passing rough a circular coil of two urns produces a magnetic field B at its center. The coil is then rewound so as to have four turns and the same current is passed through it. The magnetic field at its center now is:

A) $\frac{B}{4}$
B) 4B
C) 2B
D) $\frac{B}{2}$

Q.29 In a current carrying long solenoid the field produced does not depend on:

A) number of turns per unit length
B) current flowing
C) radius of the solenoid
D) all of these

Q.30 An electron moves with speed 2×10^5 m/s along the positive x-direction in the presence of a magnetic induction B = i + 4j + 3k (in tesla). The magnitude of the force experienced by the electron in newtons is:

A) 1.18×10^{-13}
B) 1.28×10^{-13}
C) 1.6×10^{-13}
D) 1.72×10^{-13}

Q.31 The phenomenon of production of magnetic field on passing an electric current in a straight conducting wire is based on the law of:

A) coulomb B) oersted
C) ampere D) faraday

Q.32 The magnetic induction due to a long straight conductor at a distance 2 cm from it is 10^{-6} Tesla. The current flowing in the conductor at a distance d from it is proportional to:

A) 10 A B) 1 A
C) 0.1 A D) zero

Q.33 The magnetic induction due to a current *i* passed in a straight conductor at a distance d from it is proportional to:

A) i /d B) d^2
C) i /d^2 D) d

Q.34 The magnetic field inside a solenoid is:

A) zero B) non-uniform
C) infinite D) uniform

Q.35 Which of the following quantities is not affected by magnetic field?

A) moving charge B) change in magnetic flux
C) current flowing in a conductor D) stationary charge

Q.36 A galvanometer with resistance 100 Ω gives full scale deflection with a current of 10 mA. The value of shunt, in order to convert it into an ammeter of 10 ampere range, will be:

A) 0.01 **Ω** B) 0.1 **Ω**
C) -10 **Ω** D) 1 **Ω**

Q.37 A galvanometer with resistance 5 Ω can read up to 5 mA. If this instrument is to be used to read up to 100 V, then the value of resistance to be used in its series will be:

A) 1999.5 **Ω** B) 199.995 **Ω**
C) 19.9995 **Ω** D) 19995 **Ω**

Q.38 An electric charge q is moving in a uniform magnetic field parallel to the lines of force. The magnetic force acting on the charge will be:

A) qB / v B) qvB
C) qv / B D) zero

Q.39 The length and number of turns of a solenoid are l and n respectively and the magnetic induction at the center is B. If the length of solenoid is reduced to half and the same wire is wound over it in two layers, then the new magnetic induction will be:

A) 4B B) B
C) 2B D) B / 2

Q.40 An ammeter reads up to 1A. Its internal resistance is 0.81 Ω . To increase the rage to 10 A, the value of the required shunt is:

A) 0.3 **Ω** B) 0.9 **Ω**
C) 0.09 **Ω** D) 0.03 **Ω**

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Q.41 A charged particle moving in a magnetic field experiences a resultant force:
A) In the direction of the field
B) In the opposite direction of the field
C) In the direction perpendicular to both the field and its motion
D) In the direction opposite to its motion

Q.42 The magnetic force experienced by a charged particle moving in a magnetic field will be maximum if it moves:
A) At an angle of 60° to the field
B) Parallel to the field
C) Anti-parallel to the field
D) Perpendicular to the field

Q.43 The magnetic force experienced by a charged particle moving in a magnetic field will be minimum (zero) when it moves:
A) Perpendicular to the field
B) Parallel to the field
C) Anti-parallel to the field
D) At an angle of 60° to the field

Q.44 The magnetic force acting on a unit positive charge moving perpendicular to the magnetic field with a unit velocity is called:
A) Magnetic flux
B) Electric field intensity
C) Magnetic induction
D) Self inductance

Q.45 The SI unit of magnetic induction is tesla which is also equal to:
A) Newton / ampere meter or N / A-m
B) Newton / ampere-meter^2 or N / A-m^2
C) Newton / ampere^2-meter or N / A^2-m
D) Newton / ampere^2- meter^2 or N / A-m^2

Q.46 A 5 m wire carrying a current of 2 A is at right angles to uniform magnetic field of 0.5 weber / m^2. The force on the wire is:
A) 2 N
B) 4 N
C) 5 N
D) 1.5 N

Q.47 The SI unit of magnetic flux is:
A) NmA^{-1}
B) NAm^{-1}
C) NmA^{-2}
D) Nm^{2}A^{-1}

Q.48 Magnetic flux and flux density are related by:
A) Magnetic flux = Flux density / Area
B) Magnetic flux = Flux density x Area
C) Flux density = Magnetic flux / area
D) Magnetic density = Magnetic flux x Area

Q.49 The e/m of ad electron moving with speed along a circular path in a magnetic field is give as:
A) e / m = B^{2}R / E
B) e / m = E / B^{2}R
C) e / m = E^2 / B^{2}R
D) e / m = B^2 / ER

Q.50 The magnetic field produced at a point by a current carrying straight conductor is given by:
A) B = μ_oI / 2πr
B) B = 2πr / μ_oI
C) B = 2πμ_o / Ir
D) B = Ir / 2πμ_o

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Q.51 The coil of a galvanometer is suspended in a radial field so that the deflecting torque on the coil is always:

A) NIAB cos α B) NIAB sin α
C) NIAB tan α D) NIAB

Q.52 The current passing through a coil of galvanometer is given by:

A) I = Cθ / BAN B) I = CθN / AB
C) I = ANB / Cθ D) I = AN / BCθ

Q.53 In order to increase the range of an ammeter, the shunt resistance is:

A) Decreased
B) Increased
C) Kept constant
D) Sometimes increased and sometimes decreased

Q.54 To convert a galvanometer of resistance 50 Ω and current limit 2 mA into an ammeter of range 1 ampere, the resistance required is:

A) 25 **Ω** B) 0.1 **Ω**
C) 0.2 **Ω** D) 0.05 **Ω**

Q.55 The pole pieces of the magnet in galvanometer are made concave to make the field:

A) Strong B) Weak
C) Radial only D) Radial and strong

Q.56 Such a galvanometer in which the coil comes to rest quickly after the current passed through it is called:

A) Sensitive galvanometer B) Stable galvanometer
C) Dead beat galvanometer only D) Both stable or dead beat galvanometer

Q.57 Two parallel wires carrying currents in the same direction:

A) Repel each other
B) Attract each other
C) Have no effect upon each other
D) They cancel out their individual magnetic fields

Q.58 Which one of the following relations is correct:

A) 1 Wb-m^{-2} = Nm^{-1}A^{-1} B) 1 tesla = 10^4 gausses
C) 1 Wb-m^{-2} = 1 tesla D) All of these

Q.59 The SI unit of magnetic permeability is:

A) Wb A^{-1}m^{-1} B) WbmA^{-1}
C) Wb Am^{-1} D) None of these

Q.60 Which of the following is scalar:

A) Magnetic flux B) Flux density
C) emf D) Both A & B

Q.61 The magnetic field inside the solenoid along its axis due to current is:

A) Zero B) Non- Uniform
C) Infinite line D) Strong and uniform

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Q.62 The direction of the force experienced by a charged particle moving in a magnetic field will be:
A) Parallel to the field
B) Opposite to the field
C) Parallel to its direction
D) Perpendicular to both the field and the velocity vector

Q.63 Charge to mass ration (e/m) of and electron is given by the relation:
A) $\frac{e}{m} = \frac{2V}{Br^2}$
B) $\frac{e}{m} = \frac{2V}{B^2r}$
C) $\frac{e}{m} = \frac{2V}{B^2r^2}$
D) $\frac{e}{m} = \frac{V}{2B^2r^2}$

Q.64 Tick the correct statement when the plane of the current carrying coil is at right angles to the uniform magnetic field:
A) Torque is maximum and flux is zero
B) Torque is zero and flux is maximum
C) Both Torque and flux are maximum
D) Both Torque and flux are zero

Q.65 To increase the range of an ammeter, the shunt value should be:
A) Decreased
B) Increased
C) Kept constant and R_g should be increased
D) None of these

Q.66 To convert a moving coil galvanometer into voltmeter, the series high resistance is given by:
A) $R_h = \frac{V}{I_r} + R_g$
B) $R_h = \frac{V}{I_g} - R_g$
C) $R_h = \frac{V}{R_g} - I_g$
D) $R_h = \frac{V}{R_g} + I_g$

Q.67 To increase the measuring range of a voltmeter, the series resistance should be:
A) Increased
B) Decreased
C) Kept constant and R_g should be increased
D) Zero

Q.68 Charge to mass ratio (e/m) of and electron is given by the relation:
A) $\frac{e}{m} = \frac{2V}{Br^2}$
B) $\frac{e}{m} = \frac{2V}{B^2r}$
C) $\frac{e}{m} = \frac{2V}{B^2r^2}$
D) $\frac{e}{m} = \frac{V}{2B^2r^2}$

Q.69 The force experienced by a charged particle moving in a magnetic field will be maximum when it moves:
A) Parallel to the field
B) Perpendicular to the field
C) Anti parallel to its direction of motion
D) At an angle of 45° to the field

Q.70 The magnetic force on an electron travelling 10^8 m/sec parallel to a field of strength 1 weber/m^2 is:
A) Zero
B) 10^3 N
C) 10^{-12} N
D) 16×10^{-12} N

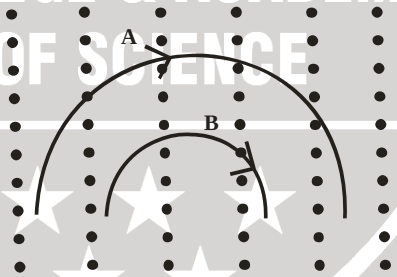
Q.71 The SI unit of flux density is:

A) Weber-m^2
B) Tesla
C) $Nm^{-1}A^{-1}$
D) All of them

Q.72 A proton travels from left to right in the plane of the paper in a magnetic field perpendicular to and directed out of the paper. It is deflected:

A) Up
B) Down
C) Into paper
D) out of paper

Q.73 Two particles A and B of masses m_A and m_B respectively and having the same charge are moving in a plane. A uniform magnetic field exists perpendicular to the plane. The speeds of the particles are v_A and v_B respectively and the trajectories are as show in figure. There:

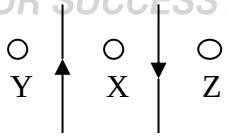


A) $m_A v_A < m_B v_B$
B) $m_A v_A > m_B v_B$
C) $m_A < m_B$ and $v_A > v_B$
D) $m_A = m_B$ and $v_A = v_B$

Q.74 An electron is accelerated by a potential difference of 12000 V. It then entrer a uniform magnetic field to 10^{-3} T applied perpendicular to the path of electron. What is the radius of path? Given mass of electron = 9 x 10^{-31} kg and charge on electron = 1.6 x 10^{-19} C.

A) 36.7 m
B) 36.74 m
C) 3.67 m
D) 3.67 cm

Q.75 Two long straight parallel wires held vertically have equal but opposite current as shown in the figure.



Which of the following effect will be observed?

A) Magnetic field at 'X' is stronger than at 'Y' and 'Z'.
C) Magnetic field at 'X','Y' and 'Z' is same.
B) Magnetic field at 'X' is weaker than that at 'Y' and 'Z'.
D) Magnetic field at 'X' is weaker than that at 'Y' but stronger than that at 'Z'.

Q.76 Due to current in a straight conductor the distance between magnetic field lines

A) Increases away from conductor
B) Decreases away from conductor
C) Increases towards conductor
D) Decreases and then increases towards conductor

ANSWERS

1.	(A)	**21.**	(C)	**41.**	(C)	**61.**	(D)
2.	(C)	**22.**	(D)	**42.**	(D)	**62.**	(D)
3.	(A)	**23.**	(B)	**43.**	(B)	**63.**	(C)
4.	(D)	**24.**	(A)	**44.**	(C)	**64.**	(B)
5.	(B)	**25.**	(B)	**45.**	(A)	**65.**	(A)
6.	(B)	**26.**	(D)	**46.**	(C)	**66.**	(B)
7.	(C)	**27.**	(C)	**47.**	(A)	**67.**	(A)
8.	(C)	**28.**	(B)	**48.**	(B)	**68.**	(C)
9.	(C)	**29.**	(C)	**49.**	(B)	**69.**	(B)
10.	(B)	**30.**	(C)	**50.**	(A)	**70.**	(A)
11.	(C)	**31.**	(B)	**51.**	(D)	**71.**	(D)
12.	(A)	**32.**	(D)	**52.**	(A)	**72.**	(B)
13.	(D)	**33.**	(A)	**53.**	(A)	**73.**	(B)
14.	(A)	**34.**	(D)	**54.**	(B)	**74.**	(B)
15.	(D)	**35.**	(D)	**55.**	(D)	**75.**	(A)
16.	(B)	**36.**	(B)	**56.**	(D)	**76.**	(B)
17.	(A)	**37.**	(D)	**57.**	(B)	**77.**	
18.	(D)	**38.**	(D)	**58.**	(D)	**78.**	
19.	(D)	**39.**	(C)	**59.**	(A)	**79.**	
20	(A)	**40.**	(C)	**60.**	(D)	**80.**	

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Solutions / Hints

5. $F = evB \sin 30^o$

7. $Bqv = \frac{mv^2}{r}$

$= \frac{mv}{Bq}$

13. $B = \mu_o nI$

Or $I = \frac{B}{\mu_o n} = \frac{20 \times 10^{-3}}{(4\pi \times 10^{-7})(20 \times 100)}$

$I \approx 8$ A

14. $\tau = BIA$

24. Parallel loops carrying currents on the same direction attract each other.

25. $F = ILB \sin 90^o$

28. Magnetic field due to circular loop at the center is given by:

$B = \frac{\mu_o NI}{2r}$

N being the number of turns in the coil.

For direction a convenient rule is a follows:

30. $\vec{v} = 2 \times 10^5\ \hat{\imath}$

$\vec{F} = q(\vec{v} \times \vec{B})$

$= (1.6 \times 10^{-19}) \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 \times 10^5 & 0 & 0 \\ 1 & 4 & -3 \end{vmatrix}$

$= (1.6 \times 10^{-19})(-6 \times 10^5\ \hat{\jmath} + 10^5\ \hat{k})$

$= 3.2 \times 10^{-14}\ [-3\hat{\jmath} + 4\hat{k}]$

Magnitude:

$F = 302 \times 10^{-14}\ (5)$

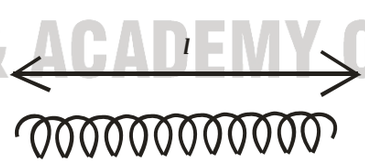
$F = 1.6 \times 10^{-13}$ N

32. $B = \left(\frac{\mu_o}{2\pi}\right)\frac{I}{r}$

37. $R = \frac{V}{I_g} - R_g$

$= \frac{100}{5 \times 10^{-3}} - 5 = 19995\ \Omega$

39.



Number of turn = n

$B = \mu_o \left(\frac{n}{l}\right) I$

If length is reduced to half, n also reduced to half then B remains constant. When same wire is wound over it in two layers, then:

$$B' = \left(\frac{2(\frac{n}{2})}{l/2}\right)I = 2B$$

73. $Bqv = \frac{mv^2}{r}$

$r = \frac{mv}{Bq}$

$r \propto mv$

Clearly → r_A r_B

$\therefore m_A v_A > m_B v_B$

74. $\frac{mv^2}{r} = Bqv$ or $r = \frac{mv}{Bq}$

$r = \frac{P}{Bq}$ or $r = \frac{\sqrt{2mE}}{Bq}$

$r = \frac{\sqrt{2mqv}}{Bq}$

$r = \frac{1}{B}\sqrt{\frac{2mv}{q}}$

$= 103\sqrt{\frac{2 \times 9 \times 10^{-31} \times 12000}{1.6 \times 10^{-19}}}$ m

$= 3 \times \sqrt{1.5} \times 10^{-1}$ m

$= 0.3 \times \sqrt{1.5} \times 100$cm

$= 36.74$ cm

ELECTROMAGNETIC INDUCTION (UNIT#09)

Q.1 According to Faraday's law of electromagnetic induction:
A) Electric field is produced by time varying magnetic flux
B) Magnetic field is produced by time-varying electric flux
C) Magnetic field is associated with a moving charge
D) none of these

Q.2 Lenz's law is a consequence of the law of conservation of:
A) charge B) current
C) momentum D) energy

Q.3 According to Faraday's law of electromagnetic induction:
A) the direction of induced current is such that it opposes the cause producing it
B) the magnitude of induced emf produced in a coil is directly proportional to the rate of change of magnetic flux
C) the direction of induced emf is such that it opposes the cause producing it
D) none of these

Q.4 A wire of length 1.0 m moves with a speed of 10 m/s perpendicular to a magnetic field. If the emf induced in the wire is 1.0 V, the magnitude of the field is:
A) 0.01 T B) 0.1 T
C) 0.2 T D) 0.02 T

Q.5 The current in a coil changes from 0 to 2 A in 0.05s. If the induced emf is 80 V, the self-inductance of the coil is:
A) 1 H B) 0.5 H
C) 1.5 H D) 2 H

Q.6 A coil having number of turns N and cross-sectional area A is rotated in a uniform magnetic field B with and angular velocity . The maximum value of the emf induced in it is:
A) $\frac{NBA}{\omega}$ B) NBA ω
C) $\frac{NBA}{\omega^2}$ D) NBA ω^2

Q.7 The mutual induction of a pair of coils is 2H. If the current in one of the coil changes from 10a to zero in 0.1 s , the emf induced in the other coil is:
A) 2 V B) 20 V
C) 0.2 V D) 200 V

Q.8 The mutual inductance of a pair of coils, each of N turns, is M henry. If a current of I ampere in one of the coils is brought to zero in t seconds, the average induced emf in the other coil, in volt, will be:

A) $\frac{MI}{t}$ B) $\frac{NMI}{t}$
C) $\frac{MN}{It}$ D) $\frac{MI}{NI}$

Q.9 An emf of 5 mV is induced in a coil when, in a nearby placed another coil, the current changes by 5 A in 0.1 s. The mutual inductance between the two coils will be:

A) 1 H B) 0.1 H
C) 0.1mH D) 0.001 mH

Q.10 Eddy currents are produced in a material when it is:

A) heated B) placed in a time varying magnetic field
C) placed in an electric field D) placed in a uniform magnetic field

Q.11 Faraday's law of electromagnetic induction is related to the:

A) law of conservation of charge
B) law of conservation of energy
C) third law of motion
D) law of conservation of angular momentum

Q.12 When a direct current *i* is passed through and inductance L the energy stored is:

A) zero B) Li
C) $\frac{1}{2}Li^2$ D) $L^2 / 2i$

Q.13 When the current in a coil changes from 2 A to 4 A in 0.05 s, and emf of 8V is induced in the coil. The coefficient of self-inductin of the coil is:

A) 0.1 H B) 0.2 H
C) 0.4 H D) 0.8 H

Q.14 The self-inductance of a coil is 5 h. A current of 1 A changes to 2 A within 5 s through the coil. The value of induced emf will be:

A) 10 V B) 0.1 V
C) 1.0 V D) 100 V

Q.15 The direction of induced emf during electromagnetic induction is given by:

A) Faraday's law B) Lenz's law
C) Maxwell's law D) Ampere's law

Q.16 To induce an emf in a coil, the linking magnetic flux:

A) must decrease B) must increase
C) must remain constant D) can either increase or decrease

Q.17 An ideal transfer is used to step up an alternating emf of 220 V to 4.4 kV to transmit 6.6 kW of power. The current rating of the secondary is:

A) 30 A B) 3 A
C) 1.5 A D) 1 A

Q.18 A current of 2 A flowing through a coil of 100 turns gives rise to a magnetic flux of 50 x 10^{-5} Wb per turn. The magnetic energy associated with the coil is:

A) 5 J B) 0.5 J
C) 0.05 J D) 0.005 J

Q.19 The current passing through a choke coil of 5 H is decreasing at the rate of 2 A/s. The emf developed across the coil is:

A) 10 V B) -10 V
C) 2.5 V D) -2.5 V

Q.20 In a step-up transformer, the turns ratio of primary and secondary is 1:2. A Laclanche cell of emf 1.5 V is connected across the primary. The voltage developed across the secondary would be:

A) zero B) 3.0 V
C) 1.5 V D) 0.75 V

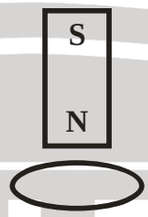
Q.21 A step-up transformer is used on a 120 V line to provide a potential difference of 2400 V. If the primary coil has 75 turns, the number of turns in the secondary coil is:

A) 1500 B) 1200
C) 150 D) 1575

Q.22 If the emf induced in a coil is 200 V when the current changes from 1 a to 2 A in 0.05s , the self-inductance of the coil is:

A) 1 H
B) 10 H
C) 20 H
D) 30 H

Q.23 The north pole of a magnet is brought near a metallic ring as shown in the figure. The direction of induced current n the ring will be:



A) anti-clockwise
B) clockwise
C) first anti-clockwise and then clockwise
D) first clockwise and then anti-clockwise

Q.24 Quantity that remains unchanged in a transformer is:

A) voltage
B) current
C) frequency
D) Both (a) & (b)

Q.25 The unit of inductance is equivalent of:

A) $\frac{\text{volt x ampere}}{\text{second}}$
B) $\frac{\text{ampere}}{\text{volt x second}}$
C) $\frac{\text{volt}}{\text{ampere x second}}$
D) $\frac{\text{volt x second}}{\text{ampere}}$

Q.26 A two meter wire is moving with a velocity of 1 m/sec perpendicular to a magnetic field of 0.5 weber/m^2. The emf induced in it will be:

A) 0.5 volt
B) 0.1 volt
C) 1 volt
D) 2 volt

Q.27 If the rotational velocity of a dynamo armature is doubled, then the induced e.m.f will be:

A) become half
B) become double
C) become quadruple
D) remain unchanged

Q.28 Energy stored in the choke coil in the form of:

A) heat
B) electricity energy
C) magnetic energy
D) electro-magnetic energy

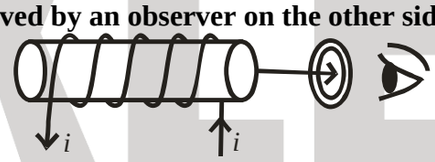
Q.29 The core of transformer is laminated so that:

A) The ratio of the voltage in the secondary to that in the primary may be increased
B) Energy losses due to eddy currents may be minimized
C) The weight of the transformer may be reduced
D) Rusting of the core may be prevented

Q.30 The core of a transformer is laminated to reduce:

A) copper loss
B) magnetic loss
C) eddy current loss
D) hysteresis loss

Q.31 A current carrying solenoid is carried towards a conducting loop. The direction of induced current in the loop as observed by an observer on the other side of the loop is:



A) towards west
B) clockwise
C) towards east
D) anti-clockwise

Q.32 The turn ratio in a step up transformer is 4:1. On passing a current of 4 A in the primary, the current in the secondary will be:

A) 0.25 A
B) 1 A
C) 2 A
D) 8 A

Q.33 When a piece of wire is passed through the space between the pole pieces of a horse shoe magnet in 0.1 s then and emf of 4 x 10^{-3} V is induced in it. The magnetic flux between the pole pieces will be:
A) 4 x 10^{-4} Wb
B) 0.1 Wb
C) 10 Wb
D) 4 x 10^{2} Wb

Q.34 The number of turns in the primary and secondary coils of a transformer are 100 and 300 respectively. If the input power is 60 W the output power will be:
A) 180 W
B) 3 x 10^{3} W
C) 60 W
D) 90 W

Q.35 An inductor may store energy in:
A) its coil
B) its magnetic field
C) its electric field
D) both in electric and magnetic field

Q.36 A 50 mH coil carries a current of 2A, the energy stored in it in Joules is:
A) 5.05
B) 0.1
C) 0.5
D) 1

Q.37 A coil of copper wire is pulled in a magnetic field with a constant velocity. If its ohmic resistance is increased then:
A) it is easier to pull it
B) it is difficult to pull it
C) it cannot be pulled
D) none of these

Q.38 The value of the current in the armature of a DC motor is maximum when the motor:
A) starts rotating with mean speed
B) has gained full speed
C) is switched off
D) just starts rotating

Q.39 Two inductance coils, of same self-inductance, L are connected in parallel and the distance between them is large. The resultant self-inductance of the coil will be:
A) L/2
B) L/4
C) 2L
D) L

Q.40 A rectangular loop of sides 8 cm and 2 cm is lying in a uniform magnetic 0.5 T with its plane normal to the field. The field is no gradually reduced at the rate of 0.02 T/sec. If the resistance of the loop is 1.6 Ω , then the power dispatched by the loop as heat is:
A) 6.4 x 10^{-10} W
B) 3.2 x 10^{-10} W
C) 6.4 x 10^{-5} W
D) 3.2 x 10^{-5} W

Q.41 According to Faraday's law of electromagnetic induction, the induced emf in a coil can be mathematically expressed as:
A) $E = -N\Delta t / \Delta\emptyset$
B) $E = -\Delta\emptyset / N\Delta t$
C) $E = -N\Delta\emptyset / \Delta t$
D) $E = -\Delta t / N\Delta\emptyset$

Q.42 Mutual inductance has a practical role in the performance of the:
A) Radio choke
B) A.C. generator
C) D.C. generator
D) Transformer

Q.43 Mutual inductance of two-coils is measured by the relation:
A) $M = -\varepsilon_s / \Delta I_p / \Delta t$
B) $M = -\varepsilon_s \Delta I_p / \Delta t$
C) $M = -\varepsilon_s / \Delta t$
D) $M = -\varepsilon_s / \Delta I_p$

Q.44 One henry can be defined as:
A) 1 henry = Amp-sec / volt
B) 1 henry = Volt-sec / Amp
C) 1 henry = Volt x Amp-sec
D) 1 henry = Volt / Amp-sec

Q.45 A 50 mH coil carries a current of 2 amps. The energy stored in its magnetic field is:
A) 0.05 J
B) 0.1 J
C) 10 J
D) 100 Joules

Q.46 The magnitude of the emf induced in the coil of A.C. generator is given by:
A) $\varepsilon = NBA \sin \omega t$
B) $\varepsilon = NBA \sin \omega t / \omega$
C) $\varepsilon = NBA\omega \sin \omega t$
D) $\varepsilon = B \sin \omega t / NA\omega$

Q.47 A dynmo converts:
A) Electrical energy into mechanical energy
B) Mechanical energy into electrical energy
C) Heat Energy into electrical energy
D) Magnetic energy into electrical energy

Q.48 A transformer is used to change:
A) The voltage of direct current
B) The voltage of alternating current
C) The electrical energy

D) The magnetic field

Q.49 If the secondary coil has N_s turns and the primary N_p turns, the relation between secondary and primary voltages is given by:

A) $V_s / V_p = N_p / N_s$ B) $V_s / V_p = N_s / N_p$
C) $V_p / V_s = N_s / N_p$ D) $V_p / V_s = N_p / N_s$

Q.50 A thin circular ring of area A is held perpendicular to a uniform magnetic field B. A small out is made in the ring and a galvanometer is connected across the ends such that the total resistance of the circuit is R. When the ring is suddenly squeeze to zero area, the charge flowing through the galvanometer is:

A) $\frac{BR}{A}$ B) $\frac{AB}{R}$
C) ABR D) $\frac{B^2A}{R^2}$

Q.51 The induced emf 'ε' produced during the phenomenon of self-inductance is given by the relation:

A) $\varepsilon_L = -L\frac{\Delta I}{\Delta t}$ B) $\varepsilon_L = -L\frac{\Delta t}{\Delta I}$
C) $\varepsilon_L = -L\,\Delta I \times \Delta t$ D) $\varepsilon_L = -\frac{\Delta I}{L\Delta t}$

Q.52 The work is stored in in the inductor as:

A) Elastic potential energy B) Kinetic energy
C) Potential energy D) Gravitation potential energy

Q.53 Energy stored per unit volume inside the solenoid is called:

A) Energy density B) Charge density
C) Mass density D) Volume charge density

Q.54 The relation for emf induced in an A.C. generator can be expressed by:

A) $\varepsilon = \varepsilon_o \cos(\omega t)$ B) $\varepsilon = \varepsilon_o \sin(\omega t)$
C) $\varepsilon = \varepsilon_o \tan(\omega t)$ D) $\varepsilon = \varepsilon_o / \sin(\omega t)$

Q.55 Slip rings are used in:

A) A.C. Dynmo B) D.C. Dynmo
C) Rectifier D) Electric Motor

Q.56 The induced emf in a motor which opposes the external emf running the motor is called:

A) Induced emf of a motor B) Back emf of a motor
C) emf D) Forward emf of battery

Q.57 If the motor is overloaded , then the magnitude of "Back emf":

A) Increases B) Decreases
C) Remains constant D) Becomes zero

Q.58 According to Faraday's law, the emf induced in a coil depends on the:

A) Maximum magnetic flux B) Rate of change of magnetic flux
C) Change in magnetic flux D) Internal magnetic flux

Q.59 The SI unit of self-induction is the same as that of:

A) Capacitance B) Mutual inductance
C) e.m.f D) Reactance

Q.60 The direction of induced current is determined by:

A) Faraday's law B) Ohm's law
C) Ampere's law D) Lenz's law

Q.61 The emf induced in the secondary coil due to rate of change of current in the primary coil is given by the relation:

A) $\varepsilon_s = -M\frac{\Delta I_p}{\Delta t}$ B) $\varepsilon_s = -\frac{\Delta I_p}{M\Delta t}$
C) $\varepsilon_s = -M\frac{\Delta t}{\Delta I_p}$ D) $\varepsilon_s = -\frac{\Delta t}{M\Delta I_o}$

Q.62 One henry is equal to:

A) 1 ohm x 1 sec B) 1 ohm x 1 meter
C) 1 ohm / 1 coulomb D) 1 ohm / 1 meter

Q.63 When the plane of the coil of and A.C. generator is perpendicular to the magnetic field, the emf produced will be:

A) Maximum B) Minimum
C) Zero D) None of these

Q.64 The value of induced emf in the rotating coil of an A.C. generator is given by:

A) $\varepsilon = N\omega AB \sin \omega t$
B) $\varepsilon = \frac{NAB}{\omega} \sin \omega t$
C) $\varepsilon = \frac{B\omega}{NA} \sin \omega t$
D) $\varepsilon = \frac{NA\omega}{B} \sin \omega t$

Q.65 For transmission of electricity to far-off places, form the A.C. power generating plant, we always use:

A) Step-down transformer
B) Step-up transformer
C) Transformer with several secondaries
D) All of these

Q.66 The energy stored per unit volume inside the solenoid is known as energy density and is given by:

A) $U_m = \frac{1}{2}\frac{B^2}{\mu_o}$
B) $U_m = \frac{2B^2}{\mu_o}$
C) $U_m = \frac{1}{2}\frac{\mu_o}{B^2}$
D) $U_m = \frac{1}{2}\mu_o B^2$

Q.67 Which one of the following is correct for a step-down transformer:

A) $N_p > N_s$
B) $N_p < N_s$
C) $N_p = N_s$
D) $N_s = 2N_p$

Q.68 Energy stored in inductor is:

A) $\frac{1}{2}mv^2$
B) $\frac{1}{2}KE^2$
C) $\frac{1}{2}LI^2$
D) $\frac{1}{2}ME^2$

Q.69 An alternating current passes through one complete cycle in 1μs, its frequency is:

A) 10^{-6} Hz
B) 10^{9} Hz
C) 10^{6} Hz
D) 10^{-3} Hz

Q.70 The correct expression of Faraday's law of electromagnetic induction is:

A) $\varepsilon = -\frac{\Delta F}{N\Delta t}$
B) $\varepsilon = -N\frac{\Delta \emptyset}{\Delta t}$
C) $\varepsilon = -N\frac{\Delta t}{\Delta \emptyset}$
D) $\varepsilon = -N\,\Delta\emptyset\Delta t$

Q.71 Transformer works on the principle of:

A) Mutual induction
B) Back emf
C) Self-induction
D) None of these

Q.72 The induced current can be increased by:

A) Using a stronger magnetic field
B) Moving the loop faster in magnetic field
C) Replacing the loop by a coil of many turns
D) All of these

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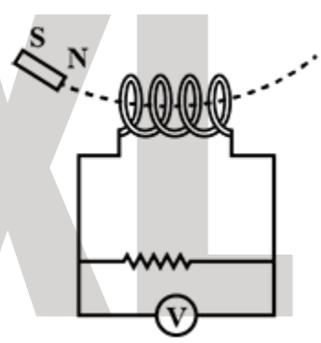
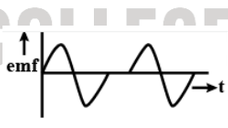
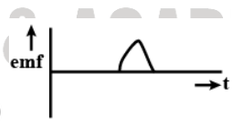
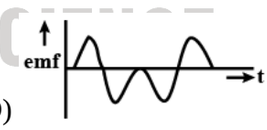
Q.73 A magnet is made to oscillate with a particular frequency, passing through a coil as shown in figure. The time variation of the magnitude of emf generated across the coil during on cycle is:



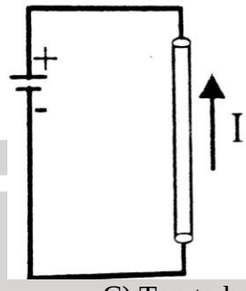
A) 

B) 

C)

D) 

Q.74 Electronic current is flowing through a straight conductor as shown in figure. The direction of magnetic lines of force will be:



A) Clockwise
B) Anticlockwise
C) Top to bottom
D) Bottom to top

ANSWERS

1.	(A)	**21.**	(A)	**41.**	(C)	**61.**	(A)
2.	(D)	**22.**	(B)	**42.**	(A)	**62.**	(A)
3.	(B)	**23.**	(A)	**43.**	(D)	**63.**	(C)
4.	(B)	**24.**	(C)	**44.**	(B)	**64.**	(A)
5.	(D)	**25.**	(D)	**45.**	(B)	**65.**	(B)
6.	(B)	**26.**	(C)	**46.**	(C)	**66.**	(A)
7.	(D)	**27.**	(B)	**47.**	(B)	**67.**	(A)
8.	(A)	**28.**	(C)	**48.**	(B)	**68.**	(C)
9.	(C)	**29.**	(B)	**49.**	(B)	**69.**	(C)
10.	(B)	**30.**	(C)	**50.**	(B)	**70.**	(B)
11.	(B)	**31.**	(D)	**51.**	(A)	**71.**	(A)
12.	(C)	**32.**	(B)	**52.**	(C)	**72.**	(D)
13.	(B)	**33.**	(A)	**53.**	(A)	**73.**	(A)
14.	(C)	**34.**	(C)	**54.**	(B)	**74.**	(A)
15.	(B)	**35.**	(B)	**55.**	(A)	**75.**	
16.	(D)	**36.**	(B)	**56.**	(B)	**76.**	
17.	(C)	**37.**	(A)	**57.**	(B)	**77.**	
18.	(D)	**38.**	(D)	**58.**	(B)	**78.**	
19.	(A)	**39.**	(A)	**59.**	(C)	**79.**	
20.	(A)	**40.**	(A)	**60.**	(D)	**80.**	

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Solutions / Hints

4. $\varepsilon = vBL$

5. $L = \frac{80}{(\frac{2}{0.05})} = 2H$

13. $L = \frac{\varepsilon}{\frac{\Delta I}{\Delta T}} = \frac{8V}{(4A-2A)/0.05sec}$

$L = 0.2\ H$

17. $I_s = \frac{P}{\varepsilon_s} = \frac{6.6}{4.4} = 1.5\ A$

18. $L = \frac{N\emptyset}{I} = \frac{100 \times 5 \times 10^{-5}}{2}$

$U = \frac{1}{2} LI^2 = \frac{1}{2} \times 2.5\ 10^{-3} \times 4$

$U = 5 \times 10^{-3}\ J$

19. $\varepsilon = L\frac{\Delta I}{\Delta t}$

$= (5H)(2A/sec)$

$= 10V$

20. A transformer does not work on a dc source.

21. $\frac{V_s}{V_p} = \frac{N_s}{N_p}$

40. Induced emf:

$= 8 \times 2\ 10^{-4} \times 0.02$

Induced current:

$= 2 \times 10^{-5}\ A$

Power loss:

$P = VI = 6.4 \times 10^{-10}\ W$

50. $\varepsilon = \frac{AB}{\Delta t}$

$q = I\Delta t$

$(\because \Delta A = A - 0 = A)$

$= \frac{AB}{R}$

$= \frac{AB}{\Delta t . R}$

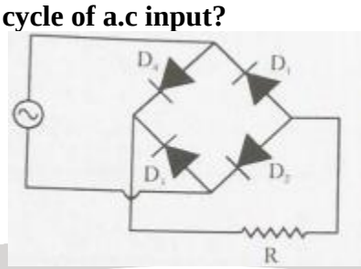
ELECTRONICS (UNIT#10)

Q.1 The similar *feature* of half wave rectifier and full wave rectifier for same input A.C source is:

A) Both produces output ripples of same frequency
B) Both uses only forward biased mode of diode
C) Both uses a pair of diodes for operation
D) Both produces pulsating D.C output

Both rectifiers produces pulsating D.C at output.

Q.2 What happens in positive cycle of a.c input?



A) D_1 and D_3 conducts
B) D_3 and D_4 conducts
C) D_1 and **D_2** conducts
D) D_2 and D_4 conducts

Q.3 The direction of the current through the load resistance of a full wave rectification circuit:
A) Inverts the negative cycle
B) changes for every cycle
C) Inverts for positive cycle
D) remains constant

Q.4 In full wave rectification bridge circuit
A) All diodes conduct simultaneously
C) Two pairs of diodes conduct alternately
B) Three diodes conduct
D) all diodes conduct one by one

Q.5 Circuit used to convert pulsating dc into constant dc is called
A) Rectifier
C) Filter
B) Inverter
D) Converter

Q.6 In half wave rectification, diode conducts for
A) Only positive half cycle of AC
C) Complete cycle of AC
B) Only negative half cycle of AC
D) quarter cycle of AC

Q.7 For a full-wave rectifier, the minimum number of diodes required is
A) 2
C) 4
B) 3
D) 1

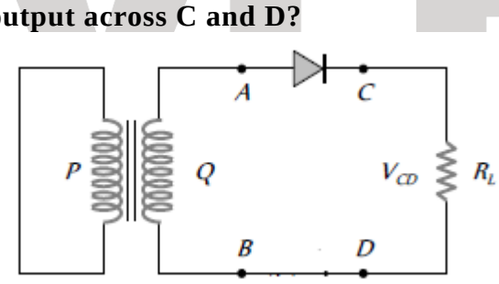
Q.8 In a full-wave rectifier, the frequency of the input AC. voltage is 50 Hz. The frequency of the output voltage is
A) 25 Hz
C) 100 Hz
B) 50 Hz
D) None of these

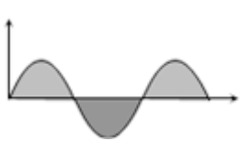
Q.9 In a HWR if Time period of input AC is 50ms, ripple frequency of the output will be:
A) 10Hz
B) 40Hz
C) 20Hz
D) 50Hz

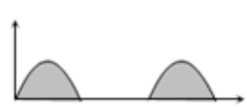
Q.10 In a HWR the load resistance is 100 Ω and input voltage is $250\sin 100\pi t$ .the value of RMS current is:
A) 5/4 A
B) 4/5A
C) 5/2 A
D) 2/5A

Q.11 In a transformer, the number of turns of primary coil and secondary coil are 5 and 4 respectively. If 220 V is applied on the primary coil, then the ratio of primary current to the secondary current is
A) 4 : 5
C) 5 : 4
B) 5 : 9
D) 9 : 5

Q.12 In the half-wave rectifier circuit shown. Which one of the following wave forms is true for V_{CD}, the output across C and D?



A) 

B) 

C)

D)

Q.13 The ratio of number of turns of primary coil to secondary coil in a transformer is 1 : 2. If a cell of 1.5 volts is connected across primary coil, the emf across secondary coil is

A) 0.75 V
C) 1.5 V
B) 0 V
D) 6 V

Q.14 An ideal transformer has 100 turns in the primary and 250 turns in the secondary. The peak value of the input ac voltage is 28 V. The r.m.s. secondary voltage is nearest to

A) 50 V
C) 70 V
B) 100 V
D) 40 V

Q.15 The primary winding of a transformer has 500 turns and its secondary has 5000 turns. If primary is connected to ac supply of 20V and 50 Hz, then secondary will have an output of

A) 2 V and 5 Hz
C) 2 V and 50 Hz
B) 200 V and 50 Hz
D) 200 V and 500 Hz

Q.16 In a step-up transformer the turn ratio is 1: 10. A resistance of 200 ohm connected across the secondary is drawing a current of 0.5 amp. What is the primary voltage and current

A) 50 V, 1 amp
C) 10 V, 5 amp
B) 25 V, 4 amp
D) 20 V, 2 amp

Q.17 In step down transformer

A) $N_p > N_s$
C) $N_p < N_s$
B) $N_p = N_s$
D) none of these

Q.18 Transformer change _____ voltage

A) A.C
C) D.C
B) Both A and C
D) none of these

Q.19 The transformer in which voltage across secondary is less than the primary voltage is called

A) Step down transformer
C) step up transformer
B) Ideal transformer
D) none of these

Q.20 In a half wave rectifier, input ac current has a frequency f, the output frequency of the current is

A) 2f
B) f/2
C) f
D) f/4

Q.21 In a full wave rectifier, input ac current has a frequency v, the output frequency of the current is

A) 2v
B) $v/2$
C) v
D) None

Q.22 In half wave rectification, for an ideal diode, if input signal is of 10V peak, 1 KHz sinusoidal, the output would be

A) 5V peak, 500Hz
B) 10V peak, 1 KHz
C) 5V peak, 1 KHz
D) 10V peak, 500 Hz

Q.23 Average voltage of half wave rectifier is approximately equals to

A) 31.80%
C) 45.60%
B) 50%
D) 23.50%

Q.24 Average voltage of half wave rectifier is

A) π x peak Value of voltage
C) π+peak Value of voltage
B) π-peak Value of voltage
D) **peak Value of voltage/π**

Q.25 In a step up transformer, the turn ratio of primary and secondary is 1 : 2. A Laclanche cell of emf 1.5 V is connected across the primary. The voltage developed across secondary would be

A) Zero
C) 3.0 V
B) 1.5 V
D) 0.075V

Q.26 The number of turns in the primary and secondary coils of a transformer are 100 and 300 respectively. If the input power is 60W. The output power will be

A) 100 W
C) 3×10^3 W
B) 60W
D) 90W

Q.27 RMS value of output of HWR is :

A) V_o
B) $V_o/\sqrt{2}$
C) $V_o/2$
D) $\sqrt{2}V_o$

Q.28 In a HWR if input Alternating voltage is $250\sin 100\pi t$, the frequency of output will be:
A) 25 Hz
B) 50 Hz
C) 100 Hz
D) None of these

Q.29 The turn ratio in a step up transformer is 4: 1 on passing a current of 4A in primary, the current in secondary will be
A) 0.25A
B) 2A
C) 1A
D) 8A

Q.30 In an ideal transformer, a bulb of 100W is operated at 220V. If primary coil has 110 turns, the power in the primary will be
A) 50W
B) 200W
C) 100W
D) Zero

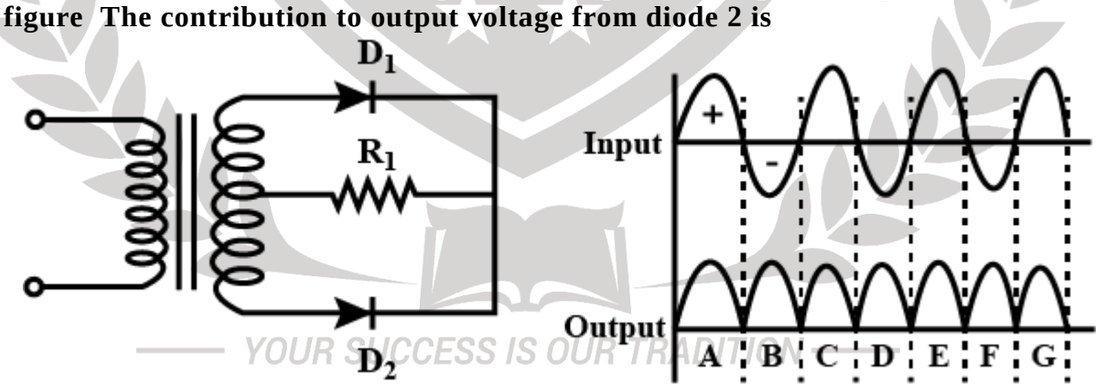
Q.31 In the case of real transformer power loss occurs due to
A) Eddy currents
B) Hysteresis loss
C) Resistance of coils
D) All of these

Q.32 The induced currents are established within the core of transformer are:
A) Along the $\vec{B}$
B) At 60° to the $\vec{B}$
C) Perpendicular to $\vec{B}$
D) At 45° to $\vec{B}$

Q.33 In a step up transformer, primary coil has 10 turns while secondary has 100 turns. A battery of emf 4.5V is connected to the primary of transformer. The voltages across secondary will be
A) 4.5V
B) 0.45V
C) 45V
D) Zero

Q.34 Efficiency of a transformer may be increased by
A) Laminating the core
B) Using copper wires of small resistance
C) Placing a soft iron piece
D) All of these

Q.35 A full wave rectifier circuit along with the input and output voltages is shown in the figure The contribution to output voltage from diode 2 is



A) A, C
B) B, C
C) B, D
D) A, D

In the positive half cycle of input ac signal diode D1 is forward biased and D2 is reverse biased so in the output voltage signal, A and C are due to D1. In negative half cycle of Input ac signal D2 conducts, hence output signals B and D are due to D2.

Q.36 In the case of real transformer
A) $P_{in} = P_{out}$
B) $P_{in} > p_{out}$
C) $P_{in} < P_{out}$
D) $P_{in} \leq p_{out}$

Q.37 A transformer is used to
A) Convert D.C in to A.C
B) Obtain required D.C voltage
C) Convert A.C into D.C
D) Obtain the required A.C voltage

Q.38 If the current in the primary coil and the number of turns in it are I_P & N_P respectively and the number of turns and current in the secondary are N_S and I_S respectively. Then the ratio N_S: N_P will be
A) $I_S : I_P$
B) $I_s^2 : I_p^2$
C) $I_P : I_S$
D) $I_P^2 : I_s^2$

Q.39 A step-up transformer is used on a 120V line to provide a potential difference of 2400V. If the primary coil has 75 turns, the number of turns in secondary coil is
A) 1500
B) 150
C) 1200
D) 1575

D) Increases the number of majority charge carriers

Q.40 If the secondary coil has N_s turns and the primary N_p turns, the relation between secondary and primary voltages is given by:

A) $V_s / V_p = N_p / N_s$
B) $V_s / V_p = N_s / N_p$
C) $V_p / V_s = N_s / N_p$
D) $V_p / V_s = N_p / N_s$

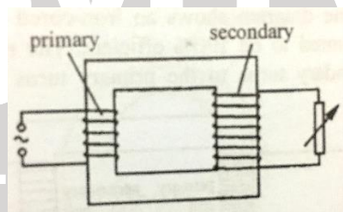
Q.41 For transmission of electricity to far-off places, form the A.C. power generating plant, we always use:

A) Step-down transformer
B) Step-up transformer
C) Transformer with several secondary
D) All of these

Q.42 Which one of the following is correct for a step-down transformer:

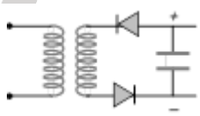
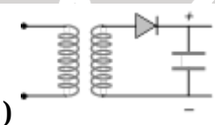
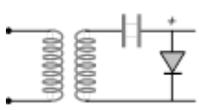
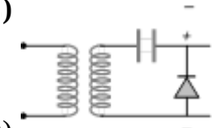
A) $N_p > N_s$
B) $N_p < N_s$
C) $N_p = N_s$
D) $N_s = 2N_p$

Q.43 Primary coil of a transformer is connected to an alternating voltage supply. The secondary coil is connected across a variable resistor. Which change will cause a decrease in the P.d across the secondary coil?

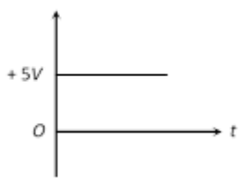
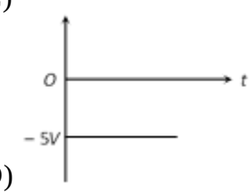


A) Increase the cross-sectional area of the secondary coil
B) Increase the current in primary coil
C) Increasing the number of turns of the primary coil
D) Increasing the resistance of the variable resistor

Q.44 Which is the correct diagram of a half-wave rectifier

A) 
C) 
B) 
D) 

Q.45 The output in the circuit of figure is taken across a capacitor. It is as shown in figure

A)
C)
B) 
D) 

Q.46 The output of full wave rectifier may be in the form of

A) Pure DC
B) Pulsating DC
C) Pulsating AC
D) A or C

Q.47 The method by which both half of A.C cycle is converted in to direct current is called

A) half wave amplification
B) half wave rectification
C) full wave rectification
D) full wave amplification

Q.48 In half-wave rectification, during negative cycle of wave, diode is
A) reverse biased
B) forward biased
C) potential barrier
D) both a and b

Q.49 Common type of rectification is
A) half wave rectification
B) full wave rectification
B) quarter wave rectification
D) both A and B

Q.50 In a half rectification, diode conducts during
A) both half cycles
B) positive half
B) negative half
C) one half input

DAWN OF MODERN PHYSICS (UNIT#11)

Q.1 The stopping potential (V_S) as a function of the frequency of the incident radiation is plotted for two different photoelectric surfaces A and B. The graphs show that work function of A is:

A) Greater than that of B
C) Smaller than that of B
B) Equal to that of B
D) No inference can be drawn about their work functions from the given graphs

Q.2 X-rays of wavelength 22 pm are scattered from a carbon target. The scattered radiation being viewed at 60° to the incident beam. What is Compton shift?
A) 1.21 Pm
B) 0.51 Pm
C) 2.43 Pm
D) None

Q.3 A photon of energy 3.4 eV is incident on a metal having work function 2 eV. The maximum kinetic energy of photoelectrons is equal to:
A) 1.4 eV
B) 1.7 eV
C) 5.4 eV
D) 6.8 eV

Q.4 The work function of a metallic surface is 5.01 eV. Photoelectrons are emitted when light of wavelength 200 A° falls on it. The potential difference required to stop the fastest photoelectrons is ($h = 4.14 \times 10^{-15}$ eVs):
A) 1.2 V
B) 2.4 V
C) 3.6 V
D) 4.8 V

Q.5 The photoelectric work function of a metal is 1 eV. Light of wavelength $\lambda = 3000$ A° falls on it. The photoelectrons will come out with approximate speed equal to:
A) 10 m/s
B) 10^2 m/s
C) 10^4 m/s
D) 10^6 m/s

Q.6 Photoelectric effect is the phenomenon in which:
A) Photons come out of a metal when it is hit by a beam of electrons
B) Photons come out of the nucleus of an atom under the action of an electric field
C) Electrons come out of a metal with a constant velocity which depends on the frequency and intensity of incident radiation
D) Electrons come out of a metal with different velocities not greater than a certain value which depends only on the frequency of the incident light and not on its intensity

Q.7 The energy of a photon of wavelength λ is:
A) $hc\lambda$
B) hc/λ
C) λ/hc
D) $h\lambda/c$

Q.8 The threshold wavelength for a photosensitive surface is 6000 A° and the wavelength of incident light is 5000 A°. Then the maximum energy of emitted electrons would be:
A) 0.041 eV
B) 0.41 eV
C) 4.1 eV
D) 41 eV

Q.9 A photon of frequency v is incident on a metal surface whose threshold frequency is v_0. The maximum kinetic energy of the emitted electron will be:

A) $h(v - v_0)$
B) $h(v + v_0)$
C) $\frac{1}{2}h(v - v_0)$
D) $\frac{1}{2}h(v + v_0)$

Q.10 The work function of a metal in 4 eV. For the emission of photoelectrons of zero velocity from the metal surface, the wavelength of the incident radiation should be:

A) 1700 A°
B) 2700 A°
C) 3100 A°
D) 5900 A°

Q.11 The maximum energy of the electrons released in a photocell is independent of:

A) frequency of incident light
B) intensity of incident light
C) nature of cathode rays
D) none of these

Q.12 In order to increase the kinetic energy of ejected photoelectrons, there should be an increase in:

A) intensity of radiation
B) wavelength of radiation
C) frequency of radiation
D) both wavelength and intensity of radiation

Q.13 Einstein's photoelectric equation is:

$$E_k = hv - \emptyset$$

In this equation E_k refers to:

A) kinetic energy of all emitted electrons
B) mean kinetic energy of emitted electrons
C) maximum kinetic energy of emitted electrons
D) minimum kinetic energy of emitted electrons

Q.14 It is necessary to consider light as a stream of photons to explain:

A) photoelectric effect
B) Compton effect
C) Polarization of light
D) diffraction of light

Q.15 The photoelectric effect is the ejection of electrons from the surface of a metal when:

A) it is heated to a high temperature
B) electrons of suitable velocity strike it
C) radiation of suitable wavelength falls on it
D) it is placed in a strong electric field

Q.16 In Compton scattering, the change in the wavelength is given by:

A) $\Delta\lambda = \frac{h}{m_o c^2}(1 + \cos\theta)$
B) $\Delta\lambda = \frac{h}{m_o c^2}(1 - \cos\theta)$
C) $\Delta\lambda = \frac{hc}{m_o}(1 + \cos\theta)$
D) $\Delta\lambda = \frac{h}{m_o c}(1 - \cos\theta)$

Q.17 A photosensitive surface is receiving light of wavelength 5000 A° at the rate of 10^{-7} J/s. The number of photons received per second is:

A) 2.5×10^{12}
B) 2.5×10^{11}
C) 2.5×10^{10}
D) 2.5×10^{9}

Q.18 Electron of mass m and charge e are accelerated through potential difference V and strike the target. The maximum speed of these electrons is

A) $\frac{eV}{m}$
C) $\frac{eV^2}{m}$
B) $\sqrt{\frac{eV}{m}}$
D) $\sqrt{\frac{2eV}{m}}$

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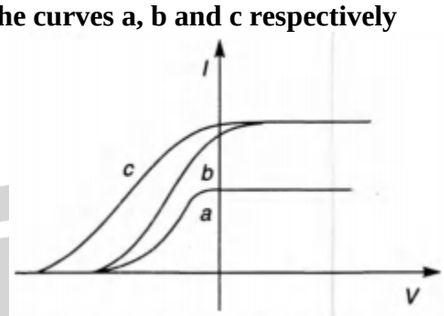
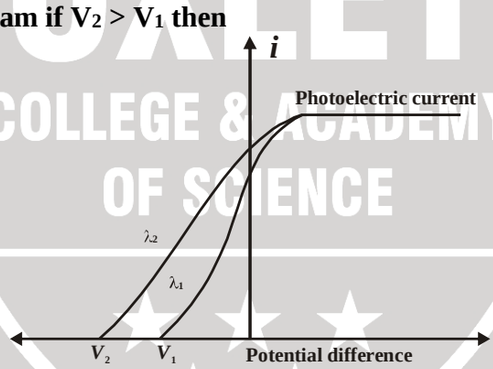
Q.19 The figure shows the variation of photocurrent with anode potential for a photosensitive surface for three different radiations. Let Ia, Ib and Ic be the intensities and fa, fb and fc be the frequencies for the curves a, b and c respectively



A) $f_a = f_b$ **and** $I_a \neq I_c$
B) $f_a = f_c$ and $I_a = I_c$
C) $f_a = f_b$ and $I_a = I_b$
D) $f_a = f_b$ and $I_a = I_b$

Q.20 In the following diagram if $V_2 > V_1$ then



A) $\lambda_1 = \sqrt{\lambda_2}$
B) $\lambda_1 = \lambda_2$
C) $\lambda_1 < \lambda_2$
D) $\lambda_1 > \lambda_2$

Q.21 A photo cell is illuminated by a small bright source placed 1 m away. When the same source of light is placed 0.5 m away, the number of electrons emitted by photo cathode would
A) Decrease by a factor of 4
B) Decrease by a factor of 2
C) Increase by a factor of 4
D) Increase by a factor of 2

Q.22 If "E" is the energy of photon, "p" is its momentum, 'f" is its frequency, "λ" is its wavelength and "c" is its speed, then which of the following relation is incorrect?
A) E = pc
B) $f = \frac{pc}{h}$
C) $p = \frac{h}{\lambda}$
D) $\lambda = \frac{h}{pc}$

Q.23 Visible region of electromagnetic radiation spectrum ranges:
A) From 400 nm to 700 nm
B) From 400 nm to 600 nm
C) From 500 nm to 750 nm
D) From 350 nm to 800 nm

Q.24 Electrons are emitted when a beam of red light falls on cathode of photocell. If the red light beam is replaced by a blue light beam of same power, which of the following quantity would decrease?
A) The work function of cathode
B) The maximum K.E of electrons emitted
C) The number of Photons striking the cathode per unit time
D) The energy of each photon striking the cathode\

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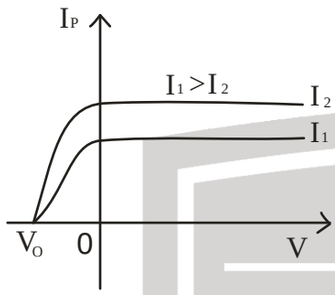
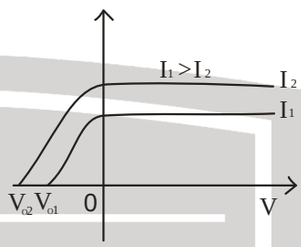
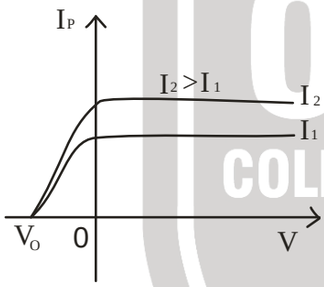
Q.25 Which of the following correctly describes the characteristics curves of photoelectric current "I_P" Vs. applied voltage for two Intensities "I_1" and "I_2" of a monochromatic light when incident on same metal:



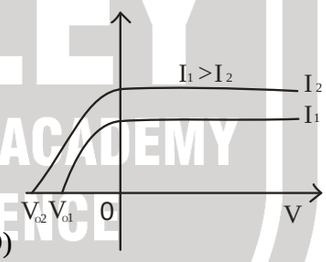
A)



B)

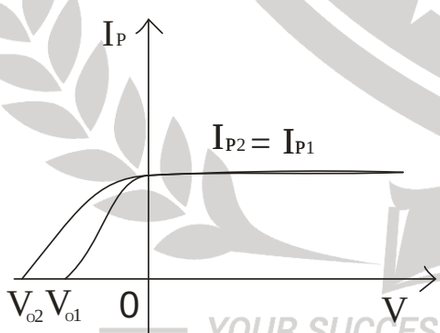


C)

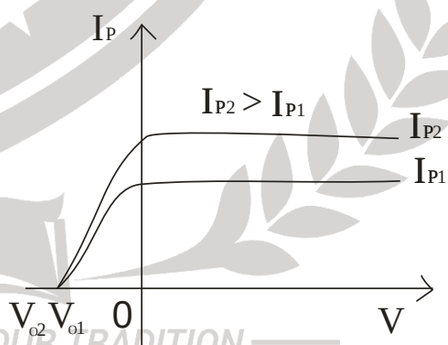


D)

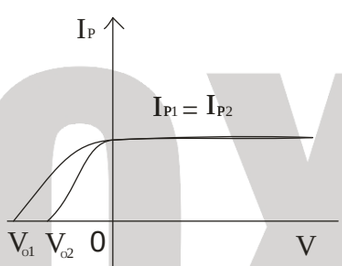
Q.26 Two light beams of same intensities and same energies are incident on two different metals with work functions "$\emptyset_2 = 2\emptyset_1$". Which of the following correctly describes the characteristics curves of photoelectric currents (I_{P1}, I_{P2}) versus, applied voltage for the given two metals:



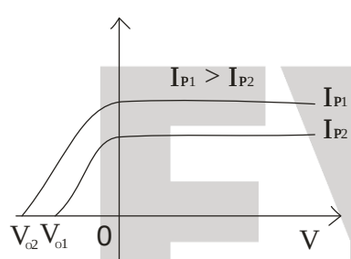
A)



B)



C)



D)

Q.27 In a photoelectric experiment a light beam of frequency greater than threshold frequency is incident on a metal. If the work function of metal is 3.5 eV and stopping potential for incident light is 5V then the maximum K.E of photoelectrons is:

A) 8.5 eV
B) 5 eV
C) 1.5 eV
D) cannot be predicted

Q.28 The amount of energy required to remove that least tightly bound electron from a metal is 2.5 eV. When a photon of some energy is incident on that metal, electros with K.Es ranging from 0 to 1.7 eV are emitted. The energy of incident photon is:

A) 0.8 eV
B) 3.7 eV
C) 4.2 eV
D) 1.9 eV

Q.29 If U.V light rays are failed to emit photoelectrons from a metal, then which of the following light can be used for the possibility of emitting the electrons from that metal?
A) Visible B) Infrared
C) Microwaves **D) X-rays**

Q.30 The threshold wavelength for a metal whose work function is 2 eV is:
A) 310 nm B) 1240 nm
C) 620 nm D) 2480 nm

Q.31 In the Einstein's photoelectric equation, K.E = hf - Ø K.E stands for:
A) K.E of all the emitted electrons **B) Maximum K.E of the emitted electrons**
C) Average K.E of the emitted electrons D) Minimum K.E of the emitted electrons

Q.32 Which of the following statement is incorrect regarding photoelectric effect?
A) Photoelectric effect displays particle nature of light
B) electrons are emitted only if frequency of light is above a threshold value
C) All emitted electrons at a particular frequency of light have same energy
D) Changing the frequency of light changes the energies of the electrons

Q.33 A source of light is placed at a distance of 100 cm from a photocell and cut off potential is found to be V_o. If the distance is double, the cut off potential will be:
A) 2 V_o B) V_o / 2
C) V_o D) V_o / 4

Q.34 The work function of a metal is 3.3 eV. The threshold frequency of that metal will be:
A) 8 x 10^{14} Hz B) 4 x 10^{14} Hz
C) 5 x 10^{20} Hz D) 2 x 10^{20} Hz

Q.35 In a photoelectric experiment, the wavelength of incident radiation is reduced from 3000 A° to 2000 A°, then:
A) Stopping potential will decrease B) K.E of emitted electrons will decrease
C) Stopping potential will increase D) Work function of metal will increase

Q.36 In a photoelectric effect experiment, the stopping potential for incident light of wavelength 500 nm is 2 V. If the wavelength is changed to 300 nm, the stopping potential is:
A) 2 V B) Less than 2 V
C) More than 2 V D) Becomes zero

Q.37 A beam of red light, a beam of yellow light a beam of green light and a beam of blue light have exactly the same energy. Which beam contains the greatest number of photons & which beam contains the smallest number of photons?
A) Green, blue B) Blue, yellow
C) Yellow, red **D) Red, blue**

Q.38 If the distance of lamp is increased from a photocell, the photoelectric current "I" in a photocell varies with distance "d" as:
A) $I \propto d^2$ **B) $I \propto \frac{1}{d^2}$**
C) $I \propto d$ D) $I \propto \frac{1}{d}$

Q.39 In a photoelectric experiment, the wavelength of incident light is increased. Which of the following is the effect of this change on the K.E of emitted electrons?
A) The average K.E decreases B) The average K.E increases
C) The maximum K.E increases D) The minimum K.E decreases

Q.40 In a photoelectric experiment, the minimum negative anode potential required to stop the electrons is 5 V. The maximum K.E of photoelectrons is:
A) 8.0 x 10^{-19} J B) 4.8 x 10^{-19} J
C) 7.6 x 10^{-19} J D) 6.4 x 10^{-19} J

Q.41 A metal surface ejects electrons when incident by blue light but no electron when incident by green light. The electrons will be ejected when the surface is incident by:
A) Red light B) Orange light
C) Violet light D) Yellow light

Q.42 The work functions fo two different metals "N" and "O" are 3.2 ev and 1.6ev respectively. If the threshold wavelength of "O" is 750 nm, then threshold wavelength of metals "N" is:
A) 375 nm B) 650 nm
C) 1500 nm D) 525 nm

Q.43 A metal is illuminated by photons of energy 1.5 eV first and then by photons of energy 3.75 eV. The ratio of maximum speed of photoelectron in two cases is_____ if the work function of metal is 0.75 eV:

A) 1 : 1 B) 1 : 4
C) 1 : 2 D) 2 : 1

Q.44 Photons of energy 6.2 eV are incident on a metal whose work function is 4.2 eV. The K.E of the fastest electrons emitted in joules is:

A) 6.8×10^{-19} **B) 3.2×10^{-19}**
C) 1.6×10^{-19} D) 4.8×10^{-19}

Q.45 Photoelectrons are emitted by a metal surface only when the light incident on it:

A) has wavelength greater than a certain minimum wavelength
B) has wavelength less than or equal to a certain maximum wavelength
C) has frequency less than or equal to a certain maximum frequency
D) Both A & C

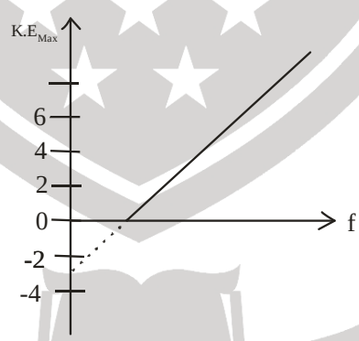
Q.46 A metal "N" is illuminated by ultraviolet and visible radiation successively and stopping potential is measured. This stopping potential is:

A) More with visible light B) Equal in both cases
C) More with ultraviolet light D) May "A" or "C"

Q.47 The photoelectric threshold for a metal is 300 nm. If the metal is illuminated with radiation of 400 nm, then the maximum K.E of emitted electrons is:

A) 3.3 eV B) 1.5 eV
C) 4.2 eV D) There is no photo emission

Q.48 The graph between maximum K.E of photoelectrons versus. Frequency of incident photon on a metal surface is shown as:



The work function of the metal is:

A) 1.5 eV B) 4.5 eV
C) 3.0 eV D) can't be predicted

Q.49 Which of the following is correct about photoelectric effect?

A) The electrons are emitted if frequency of incident photons is below a critical value
B) The electrons are emitted if wavelength is more than a critical value
C) The K.E of photoelectrons is directly proportional to intensity of incident photon beam
D) The spread of photoelectrons is inversely proportional to work function

Q.50 The threshold frequency of a metal is 3×10^{14} Hz. The work function for that metal is:

A) 1.0×10^{-19} J B) 2.7×10^{-19} J
C) 4.8×10^{-19} J **D) 2.0×10^{-19} J**

ATOMIC SPECTRA & NUCLEAR PHYSICS (UNIT#12+13)

Q.1 The half-life of an element is 160 days. What amount of it will be left in 80 days?

A) 1/8 B) 1/4
C) 1/2 **D) $1/\sqrt{2}$**

Q.2 A radioactive material has an initial amount 32g. After 60 days it reduces to 2 g, then the half-life of radioactivity material is:

A) 8 days B) 30 days
C) 60 days **D) 15 days**

Q.3 Certain radioactive substance reduces to 25% of its value in 16 days. Its half-life is:

A) 8 days B) 28 days
C) 32 days D) 64 days

Q.4 Half-life of radium is about 1600 years. In how many years shall the earth loose al its radium due to radioactive decay?
A) 1590 x 10^6 years
B) 1590 x 10^{24} years
C) 1590 x 10^{12} years
D) Never

Q.5 The Half-life of radium is about 1600 years. If 100g radium existing now, 25g will remain un decayed after:
A) 4800 years
B) 6400 years
C) 6400 years
D) 3200 years

Q.6 A radioactive substance has a half-life of four months. Three – four of the substance will decay in:
A) Three months
B) Four months
C) Eight months
D) Twelve Months

Q.7 In 420 Days, the activity of a sample of polonium fell to 1/8 th of its initial value. The Half-life of polonium is:
A) 140 days
B) 280 days
C) 70 days
D) 210 days

Q.8 Half-life of a radioactive substance is how much percent of its mean life?
A) 35%
B) 50%
C) 70%
D) 85%

Q.9 The half-life period of a radioactive nuclide is 3 hours. In 9 hours, its activity will be reduced by a factor of:
A) 1/27
B) 1/8
C) 1/6
D) 1/9

Q.10 280 days old radioactive sample has activity 6000 dps (disintegration per second). After 140 days activity decrease to 3000 dps. The initial activity in dps was:
A) 24000
B) 12000
C) 16000
D) 9000

Q.11 The Half-life of radium is 1600 years. What fraction of a sample of radium will be disintegrated after 6400 years?
A) 7/8
B) 15/16
C) 1/16
D) 1/8

Q.12 The relationship between decay constant γ and half-life $T_{1/2}$ of the radioactive substance is:
A) $\lambda = \frac{0.693}{T_{\frac{1}{2}}}$
B) $\lambda = \frac{T_{\frac{1}{2}}}{0.693}$
C) $\lambda = -\frac{ln\frac{1}{2}}{T_{\frac{1}{2}}}$
D) Both A & B

Q.13 SI unit of radioactivity is:
A) Curie
B) Becquerel
C) Roentgen
D) Rutherford

Q.14 The decay constant λ of a radioactive sample:
A) Decreases as the age of atoms increase
B) Increases as the age of atoms increase
C) Is independent of the age of atom
D) Depends on the nature of activity

Q.15 Half-life of a radioactive substance depends upon?
A) Temperature
B) Pressure
C) Nature of substance
D) Electric and magnetic field

Q.16 Decay of one radioactive nucleus per second is equal to:
A) One curei
B) One Becquerel
C) One half-life
D) One Henry

Q.17 What percentage of radioactive substance is left after five half-lives:
A) 31 %
B) 0.3 %
C) 3.125 %
D) 1 %

Q.18 Hydrogen atom can give spectral lines in the series named as Lyman, Pfund, Bracket, Balmer and Panchen. Which of the following is correct?
A) Paschen Series is in visible region
B) Bracket series is in infrared region
C) Pfund series is in ultraviolet region
D) Balmer series is in X-rays region

Q.19 Infrared region (to which three spectral series of the hydrogen spectrum belong) lies between:
A) Radio waves and microwaves
B) Visible region and ultraviolet region
C) Microwaves and visible region
D) Ultraviolet region and X-rays

Q.20 Which of the following correctly represents the ground state and all the possible excited states for paschen series of the hydrogen atom?
A) n = 1, n = 2, to ∞
B) n = 3, n = 4, to ∞
C) n = 2, n = 3, to ∞
D) n = 4, n = 5, to ∞

Q.21 A stream of electrons accelerated from rest through a petunia difference of 12.75 V is bombarded on monoatomic hydrogen gas. In the emission spectrum one cannot observe any line of:
A) Lyman series
B) Balmer series
C) Paschen series
D) Pfund series

Q.22 An electron jumps from 4th orbit to 2nd orbit. The wavelength of emitted radiation is:
A) $\frac{16}{R_H}$
B) $\frac{16}{5R_H}$
C) $\frac{16}{3R_H}$
D) $\frac{16}{7R_H}$

Q.23 The energies required in electron volt to remove an electron from three lowest orbits of hydrogen atom are:
A) 13.6, 3.4, 1.5
B) 13.6, 6.8, 3.7
C) 13.6, 10.2, 3.4
D) 13.6, 9.5, 4.3

Q.24 Which series of hydrogen atom was first discovered?
A) Lyman
B) Brackett
C) Balmer
D) Pfund

Q.25 A hydrogen atom makes a transition from 3rd excited state to 1st excited state, the energy of the photon emitted is:
A) 12.75 eV
B) 2.55 eV
C) 12.08 eV
D) 1.51 eV

Q.26 The difference in angular momentum associated with the electron in the two successive orbits of the hydrogen atoms are:
A) $\frac{h}{\pi}$
B) $\frac{h}{2}$
C) $\frac{h}{2\pi}$
D) $\frac{(n-1)h}{2\pi}$

Q.27 As the quantum number increase, the energy difference between consecutive energy levels of an atom:
A) Increases
B) Remains same
C) Decreases
D) becomes infinite

Q.28 The energy of the ground electronic state of hydrogen atom is -13.6 eV. The energy of 2nd excited state is:
A) – 3.40 eV
B) – 0.85 eV
C) – 1.51 eV
D) – 0.54 eV

Q.29 When a hydrogen atom is raised from ground state to an excited state:
A) P.E increases and K.E decreases
B) Both K.E and P.E increases
C) P.E decreases and K.E increases
D) Both K.E and P.E decreases

Q.30 Band spectrum is produced by:
A) He
B) H
C) H_2
D) Na

Q.31 According to Bohr's Model of Hydrogen atom:
A) The linear velocity of electrons is quantized
B) The total linear momentum of electrons is quantized
C) The angular momentum of electrons is quantized
D) The angular velocity of electrons is quantized

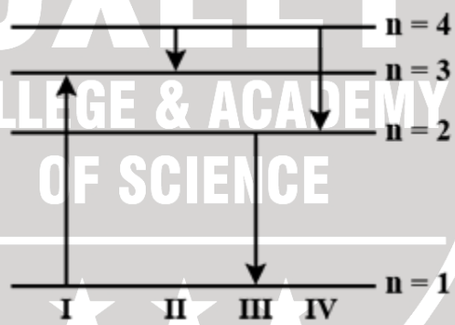
Q.32 The minimum wavelength of the Lyman series for the Hydrogen atom is given by:
A) 4 / R
B) 9 / R
C) 16 / R
D) 1 / R

Q.33 The energy of an electron:
A) is greater in outer orbit than I inner orbit
B) Is always the same whichever the orbit may be
C) Decreases as the quantum number increases
D) is greater in inner orbits than in outer orbits

Q.34 Every series of Hydrogen Spectrum has an upper & lower limit in wavelength. The Spectral series which has an upper limit of wavelength equal to 18750 A º is:
A) Lyman series
B) Balmer series
C) Pfund series
D) Paschen series

Q.35 It is known that ionization potential of hydrogen is 13.6 eV. When electron jumps from the first excited state to ground state, the emitted photon will have energy of:
A) 3.4 eV
B) 13.6 eV
C) 10.2 eV
D) 1.51 ev

Q.36 The diagram shows the energy levels for an electron in a certain atom. Which transition shown represents the emission of a photon with the most energy?



A) I
B) II
C) III
D) IV

Q.37 The energy of the highest energy photon of the Balmer series of hydrogen spectrum is close to:
A) 3.4 eV
B) 1.5 eV
C) 0.85 eV
D) 13.6 eV

Q.38 The minimum wavelength for the Lyman series of the hydrogen atom is "X", then the minimum wavelength for the Balmer series of the hydrogen atom is:
A) X
B) 2X
C) 4X
D) X / 2

Q.39 Which of the following transition in the hydrogen atom gives and emission of highest frequency?
A) n = 2 to n = 1
B) n = 4 to n = 3
C) n = 3 to n = 2
D) n = 5 to n = 2

Q.40 The series limit wavelength (minimum wavelength) of bracket and Pfund series are respectively:
A) $\frac{1}{R_H}, \frac{9}{R_H}$
B) $\frac{16}{R_H}, \frac{25}{R_H}$
C) $\frac{4}{R_H}, \frac{7}{R_H}$
D) $\frac{9}{R_H}, \frac{13}{R_H}$

Q.41 When the electron in the hydrogen atom jumps from 2nd orbit to 1st orbit, the wavelength of emitted radiation is λ. When an electron jumps from 3rd orbit to 1st orbit, the wavelength of emitted radiation is:
A) $\frac{27}{32}\lambda$
B) $\frac{25}{13}\lambda$
C) $\frac{32}{27}\lambda$
D) $\frac{13}{25}\lambda$

Q.42 The angular momentum of electron in nth orbit is:
A) nh
B) $\frac{nh}{2\pi}$
C) $\frac{2\pi}{nh}$
D) $\frac{n^2h}{2\pi}$

Q.43 The ratio of the energies of the hydrogen atom in first second excited state is:

A) $\frac{4}{9}$ B) $\frac{1}{4}$

C) $\frac{9}{4}$ D) $\frac{4}{1}$

Q.44 The ionization energy of hydrogen atom is 13.6 eV. The energy require to remove an electron from 2nd orbit is:

A) 13.6 eV B) 9.5 eV

C) 3.4 eV D) 1.5 eV

Q.45 Which of the following series of the hydrogen atom possess photons of longest wavelength?

A) Lyman B) Bracket

C) Balmer **D) Pfund**

Q.46 The diagram shown represents energy levels of a certain atom. When atom moves from "4E" energy level to "E" energy level a photon of wavelength "λ" is emitted. What will be wavelength of photon when atom moves from "$\frac{8E}{3}$" to "E":

A) $\frac{7}{5}\lambda$ **B) $\frac{9}{5}\lambda$**

C) $\frac{3}{2}\lambda$ D) $\frac{7}{4}\lambda$

Q.47 According to Bohr's Atomic Model:

A) An atom has heavy, negatively charged nucleus

B) The electron radiate energy only when it jumps to inner orbit

C) The electron can move only in particular orbit

D) Both B & C

Q.48 Ionization energy of Hydrogen atom is 13.6 eV. Hydrogen atoms in the ground state are excited by monochromatic radiation of photon of energy 12.1 eV. The No. of spectral lines emitted by hydrogen atoms according to Bohr's theory will be:

A) One **B) Three**

C) Two D) Four

Q.49 The ionization potential of an atom in ground state is 16 V. Its first excitation energy is:

A) 8 eV **B) 12 eV**

C) 4 eV D) 16 eV

Q.50 The Lyman transition involve:

A) Largest changes of potential energy B) Smallest changes of potential energy

C) Smallest changes of total energy **D) Largest changes of total energy**

Q.51 The energy required to remove and electron in a hydrogen atom from n=10 state is:

A) 0.136 eV B) 1.36 eV

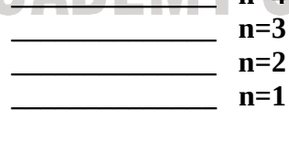
C) 13.6 eV D) 0.0136 eV

Q.52 In the Spectrum of hydrogen atom, the ratio of the longest wavelength in the Lyman series to the longest wavelength in the Balmer series:

A) 9 / 5 B) 3 / 2

C) 5 / 27 D) 4 / 9

Q.53 Four lowest energy level s of H atom are shown in figure, the No. of possible emission lines (if electron is initially in state n=4) would be:



A) 4 B) 5

C) 3 **D) 6**

Q.54 Decay constant λ given by

A) $-\frac{\Delta N / N}{\Delta t}$ C) –NΔN/Δt

B) $\frac{N}{\Delta N} / \Delta t$ D) $-\frac{\Delta t}{\Delta N / N}$

Q.55 The half-life of ^{239}U in minutes

A) 4.5×10^{9} **C) 23.4**

B) 3.8 D) none of them

Q.56 A substance reduces to one sixteenth of its original mass in 2 hours. The half-life period of the substance will be

A) 30 min C) 45min

B) 90min D) 60min

Q.57 In radioactivity, the rate of decay

A) Can be increased by magnetic field C) Can be decreased by the field

B) Can be kept constant by the electric field

D) Is not effected by electric and magnetic field

Q.58 Different radioactive materials have

A) Same half-life C) different half-life

B) Same total life **D) both B & C**

Q.59 If $\frac{15}{16}$ of radioactive atoms decays in 16hrs, the half-life of elements is

A) 16 hrs. **C) 4 hrs.**

B) 8 hrs. D) 2 hrs.

Q.60 $\lambda \times T_{1/2}$

A) 0.963 **C) $\log_e 2$**

B) $\log_e 1/2$ D) A and C

Q.61 With reference to the interaction of electrons (cathode) with atoms of the anode, what percentage of typically heat occurs

A) 1% C) 0.1%

B) 10% D) 99%

Q.62 Longest wavelength of radiation for the Paschen series is:

A) $\frac{7R}{144}$ C) $\frac{5R}{36}$

B) $\frac{R}{4}$ D) $\frac{11R}{98}$

Q.63 Find the wavelength of the spectral line corresponding to the transition in hydrogen from n = 6 state to n = 3 state?

A) $\frac{7R}{144}$ **C)** $\frac{5R}{36}$

B) $\frac{R}{6}$ **D)** $\frac{3R}{36}$

Q.64 Fraction of radioactive substance left undecayed after n half-lives is:

A) $\left(\frac{1}{2}\right)^{n}$ B) $\left(1-\frac{1}{2}\right)^{n}$ C) $\left(1-\frac{1}{2}\right)^{1/n}$ D) $\left(\frac{1}{2}\right)^{1/n}$

Q.65 It is a general principle for all radioactive element that whatever may be their half-life, their full life will be.

A) Finite **B) Infinite** C) Constant D) None of these

Q.66 The relation between half-life and decay constant is

A) $\lambda T_{1/2} = \log_e^2$ B) $\lambda T_{1/2} = \log_2$ C) $\lambda = 0.693 T_{1/2}$ D) $T_{1/2} = 0.993\lambda$

Q.67 The half-life of $^{91}_{38}Sr$ is 9.70 hours. The value of its decay constant will be

A) 1.99×10^{-5} s B) 9.11×10^{-4} s C) 2.99×10^{-4} s D) 2.94×10^{-4} s

Q.68 The half-life of a radioactive element is such that $7/8$ of a quantity of it decays in 12 days what fraction of it remain undecayed after 24 day?

A) 0 B) $\frac{1}{128}$ **C) $\frac{1}{64}$** D) $\frac{1}{32}$

Q.69 What percentage of the parent isotope remains after 0.5 half-lives have passed?

A) 70.7% B) 100% C) 50% D) 0.5%

Solution:

$(1/2)^n$ = decimal amount remaining

Where n = the number of half-lives

$(1/2)^{0.5} = 0.707$

The question asks for a percentage, so 70.7%

Q.70 The decay of a radioactive nuclide is represented by the equation $\frac{dN}{dt} = -\lambda N$ where $\lambda = 2.4 \times 10^{-8} s^{-1}$. The half-life of the nuclide will be

A) 2.9×10^{7}s B) 3.4×10^{-8}s **C) 1.3×10^{7}s** D) 8×10^{-8}s

Q.71 Sodium-24 has half-life of 15 hours and it is used in medicine estimate:

A) Kidney function C) Iron in plasma

B) Plasma blood volume D) Thyroid function

Q.72 What is the absorbed dose 'D' of a sample of 2Kg which is given an amount of 100J of radioactive energy:

A) 200Gy B) 102 Gy

C) 50 Gy D) 98 Gy

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